

Maternal Health and Well-Being

Samuel Akombeng Ojong, UNICEF Democratic Republic of Congo, Bridgette Wamakima, Harvard Medical School, Cheryl A. Moyer, University of Michigan, Ann Arbor, and Marleen Temmerman, Aga Khan University

<https://doi.org/10.1093/acrefore/9780190632366.013.214>

Published online: 18 October 2023

Summary

Maternal health and well-being refers to the physical, psychological, and emotional well-being of women during pregnancy, childbirth, and the postnatal period, as well as the absence of any morbidities or death either due to pregnancy or its management.

Despite making a comparatively late appearance on the international global policy agenda, maternal health and well-being has progressively become a global health policy priority following Deborah Maine's revolutionary article on maternal mortality. Consequently, key international policy events from Alma Ata to the International Conference on Population and Development events, through the Millennium Development Goals to the Sustainable Development Goals (SDGs) in the last decade have consecrated women's inalienable right to safe and respectful health services. Also, the growing focus on rights-based care against the backdrop of the need to ensure equity in all communities worldwide has led to an evolution in policy focus, calling on health systems to not only protect women and girls from preventable deaths but to also empower them to thrive, all while recognizing their unique role is ensuring the positive transformation of the communities in which they live.

This increasing policy attention has contributed to a disproportionate yet marked reduction in global maternal mortality and morbidity statistics over the last 30 years. However, if the world is to achieve its 2030 SDGs women's health and gender equality agendas, it is important to recognize that the broad concept of women's health cannot be limited to the rather narrow window of pregnancy, childbirth, and the postpartum period. While there are huge gaps in all resource-type settings in promoting and protecting women's agency and autonomy, the fact remains that in addition to ensuring the availability of and access to high-quality maternal health services, women's health outcomes are inextricably linked to their decision-making power on key issues such as when to become sexually active, the use of contraception, whether or not they want to achieve pregnancy and childbirth, and access to safe abortion care services. Additionally, the growing burden of noncommunicable diseases and the increasing occurrence of worldwide pandemics are providing novel challenges to the health and well-being of the world's most vulnerable women and girls, thus creating the need to ensure resilient health systems that are considerate of the rights and wishes of the world's women and girls.

Keywords: maternal health, autonomy, women's agency, maternal mortality, severe maternal morbidity, family planning, preconception care, antenatal care, delivery, postpartum

Subjects: Sexual and Reproductive Health, Special Populations

Introduction

The World Health Organization (WHO) defines *maternal health and well-being* as the physical, psychological, and emotional well-being of women during pregnancy, childbirth, and the postnatal period (World Health Organization [WHO], 2022b). This includes the absence of any long- or short-term complications during pregnancy and delivery, or the management of these (morbidity), as well as the absence of death (mortality) during these periods and up to six weeks postpartum (Koblinsky et al., 2016; Miller et al., 2016).

The WHO's definition of maternal health suits the goal-oriented approach of the Millennium Development Goals (MDGs) and the more ambitious Sustainable Development Goals (SDGs). However, it has attracted criticism from those who think that focusing on pregnancy and birth-related risks does not sufficiently address the broader concept of women's health and inadvertently ignores the role women play as key developmental actors in all communities (Knaul et al., 2016; Langer et al., 2015).

Improving maternal health was an explicit goal of the MDGs, and between 1990 and 2015, the maternal health approach guided by MDG 5 led to a 44% decline in maternal deaths worldwide (Bongaarts, 2016). Nonetheless, there were significant global variations in this decline, with sub-Saharan Africa remaining disproportionately affected (Alkema et al., 2016; Kassebaum et al., 2014). At the same time, the need to look beyond survival and investigate the capacity of countries to offer high-quality maternal healthcare services arose. Assessment of healthcare quality prompted discussions about the need to adopt a human rights lens in the delivery of these services, as well as to address issues of equity, social justice, and respectful maternity care (Tunçalp et al., 2015).

Despite the burden of pregnancy-related morbidity and mortality globally, increasingly more women are dying from noncommunicable diseases (including cervical, breast, and ovarian cancer) than from pregnancy-related conditions (44% vs. 7%), even in high maternal mortality burden regions like sub-Saharan Africa (Bongaarts, 2016; Kassebaum et al., 2014; Norheim et al., 2015). In addition to these new trends, disease outbreaks and epidemics such as Zika, Ebola, and more recently COVID-19 have emphasized the necessity to prioritize health system resilience to ensure the continued provision of high-quality maternal and women's health services. These outbreaks have also highlighted the need for innovative service delivery structures as well as the possibility of reconfiguring the current patterns of maternal healthcare service delivery (Lucas & Bamber, 2021; Wu et al., 2020).

The fight against preventable maternal mortality and morbidity remains a key guiding principle in the efforts aimed at improving access to care for women and adolescent girls worldwide, especially in resource-limited settings. However, the first part of the 21st century has ushered in a shift toward a more person-centered and rights-based approach to addressing maternal health issues. Beyond survival and the need to eliminate preventable deaths, the maternal health policy community has called and continues to call for thriving and for transforming health systems to adopt a more holistic approach to women's health and well-being. This approach must enable

women and girls to not just survive but also to thrive in enabling environments, with the maternal health policy community themselves playing a key role in transforming their communities (Brizuela & Tunçalp, 2017; Khosla et al., 2015; Lusti-Narasimhan et al., 2014; Miller et al., 2016; Temmerman et al., 2015; Tunçalp et al., 2017; WHO, 2018a).

Maternal health is also situated within a broader movement focused on improving gender equity. In what has been described as an unparalleled demonstration of global consensus, MDGs 3 and 5 specifically identify the need to “promote gender equality and empower women” and “improve maternal health,” respectively, as key components of eradicating poverty globally (WHO, 2018b, p. 16). This show of global unity yielded some remarkable results, with a close to a 44% reduction in global maternal mortality (Alkema et al., 2016; Beattie et al., 2015; Nour, 2014). Despite falling short of the 75% goal, the MDG progress reports demonstrated that drastic changes are possible and made a strong case for investing in women’s health services, including family planning, and paved the way for the SDGs by reinforcing global resolve to prioritize the health and human rights of women and girls worldwide (Beattie et al., 2015; Nour, 2014; United Nations Development Programme, 2015; Waage et al., 2010).

The SDGs therefore sought to build on the successes and lessons learned from the MDGs, identifying women’s health and gender equality as key objectives. In addition to SDG 3, which focuses on improving health (including an explicit goal to reduce global maternal mortality to a ratio of less than 70 maternal deaths per 100,000 live births), SDG 5 focuses on advancing gender equity (see Table 1 and Table 2) and includes specific targets focused on ensuring universal access to sexual and reproductive health and reducing harmful practices such as child marriage and female circumcision (Sustainable Development Goals [SDG], 2022). Underpinning discussions of gender equity are such concepts as agency, empowerment, and autonomy, none of which has a universally accepted definition or measurement strategy. Thus, the landscape of maternal health is complex, even before delving into specific challenges and opportunities.

This article therefore examines key aspects of maternal health and provides information on the relevance of maternal health within the global health landscape, the determinants of maternal health and well-being, the quality of and access to maternal healthcare services, the uptake of key maternal health services, and women’s agency, including their reproductive health rights. Each section highlights the contrasting maternal health experiences between low- and middle-income countries (LMICs) and high-income countries (HICs), while identifying key challenges, opportunities, and gaps (especially in sub-Saharan Africa).

Table 1. United Nations Sustainable Development Goal 3 (Good Health and Well-Being) Targets Related to Maternal Health

3.1 By 2030, reduce the global maternal mortality ratio to less than 70 per 100,000 live births.
3.2 By 2030, end preventable deaths of newborns and children under 5 years of age, with all countries aiming to reduce neonatal mortality to at least as low as 12 per 1,000 live births and under-5 mortality to at least as low as 25 per 1,000 live births.

3.7 By 2030, **ensure universal access to sexual and reproductive health-care services**, including for family planning, information and education, and the integration of reproductive health into national strategies and programs.

3.8 Achieve universal health coverage, including financial risk protection, access to quality essential health-care services and access to safe, effective, quality, and affordable essential medicines and vaccines for all.

Table 2. United Nations Sustainable Development Goal 5 (Gender Equality) Targets

5.1 End all forms of discrimination against all women and girls everywhere.

5.2 Eliminate all forms of violence against all women and girls in the public and private spheres, including trafficking and sexual and other types of exploitation.

5.3 Eliminate all harmful practices, such as child, early and forced marriage and female genital mutilation.

5.4 Recognize and value unpaid care and domestic work through the provision of public services, infrastructure and social protection policies and the promotion of shared responsibility within the household and the family as nationally appropriate.

5.5 Ensure women's full and effective participation and equal opportunities for leadership at all levels of decision making in political, economic and public life.

5.6 Ensure universal access to sexual and reproductive health and reproductive rights as agreed in accordance with the Programme of Action of the International Conference on Population and Development and the Beijing Platform for Action and the outcome documents of their review conferences.

5.A Undertake reforms to give women equal rights to economic resources, as well as access to ownership and control over land and other forms of property, financial services, inheritance, and natural resources, in accordance with national laws.

5.B Enhance the use of enabling technology, in particular information and communications technology, to promote the empowerment of women.

5.C Adopt and strengthen sound policies and enforceable legislation for the promotion of gender equality and the empowerment of all women and girls at all levels.

Maternal Health: Why Engage in the Discourse?

The discussion about maternal health within the global health space is important for many reasons. The universal right to health and well-being for everyone (including women and girls) argues for the need to adopt a rights-based lens in all processes that analyze the pertinence, delivery, and effectiveness of maternal health interventions and policies, while concomitantly

recognizing the inherent complexities of adopting such an approach. Furthermore, the sheer demand for essential maternal health services worldwide, and the pressure this demand exerts on health systems, illustrates the global importance of maternal health and the need for countries to enlist it among their health priorities. Women play a crucial role in national, regional, and global development, and consequently it is important to make a compelling case for ensuring the provision of high-quality, integrated sexual and reproductive health services (including maternal health services) to ensure better health, socioeconomic, and environmental outcomes for women (including adolescent girls), newborns, and children.

International Global Policy Priority

Maternal health (and pregnancy-related conditions) gained recognition on the global health policy agenda following Rosenfield and Maine's revolutionary article on maternal mortality (Shiffman & Smith, 2007; Weil & Fernandez, 1999). Consequently, maternal health has been increasingly prioritized, as demonstrated by multiple national, regional, and global initiatives. Global initiatives that have dedicated key policy statements and strategies to improving the health and well-being of women worldwide include Alma Ata, the 1994 and 2019 International Conferences on Population and Development, the Millennium Development Goals in 2000, Family Planning 2020 and 2030, and the Protocol to the African Charter on Human and Peoples' Rights on the Rights of Women in Africa ("Cairo + 25," 2019; Hardee & Jordan, 2021; Okonofua, 2005; Sommer & Forman-Rabinovici, 2019; Thompson et al., 2018).

Since 2015, United Nations Sustainable Development Goals (SDGs) have emphasized the importance of maternal health and well-being by aiming to achieve a maximum of 70 maternal deaths per 100,000 livebirths, and 12 neonatal deaths per 1,000 livebirths worldwide by 2030 (SDG 3). In addition to addressing maternal and newborn mortality and morbidity, this target further addresses the impact of epidemics such as HIV, COVID-19, Ebola, Zika virus, and monkeypox, as well as noncommunicable diseases, which are changing the configuration of the demand and supply of maternal health services worldwide (Cusimano et al., 2014; Kotlar et al., 2021; Lucas & Bamber, 2021; Smith, 2015). In addition to advocating for access to quality maternity care services throughout pregnancy, this increased global prioritization has drawn attention to the need to address major drivers of adverse health outcomes for women and girls, such as unmet need for contraceptives and high rates of unsafe abortions (Cheng et al., 2008; Lusti-Narasimhan et al., 2014; Rodríguez et al., 2014; Say et al., 2014; Temmerman, 2016, 2019).

Maternal Health and Human Rights

The persistent burden of maternal and neonatal mortality and morbidity worldwide, and the disparities in this burden between low- and high-income countries (HICs), as well as the disparities across different groups within the same population, reinforces the need to identify and address the root causes of inequalities in maternal health and well-being. The right to health and well-being for all, as contained in the SDGs, incorporates women's universal right to access

respectful, high-quality, integrated care before and during pregnancy, delivery, and the puerperium (Khosla et al., 2015; Starrs et al., 2018; Temmerman et al., 2014). However, multiple factors continue to challenge these rights.

In many low- and middle-income countries (LMICs), the absence of essential antenatal and postnatal care services, the absence of or poor uptake of comprehensive family planning services, the sheer number of unsafe abortions with poor-quality post-abortion care, and the unavailability of facility-based and/or skilled delivery care (including untimely care) contribute to the high burden of adverse maternal and newborn health outcomes (Brizuela & Tunçalp, 2017; Koblinsky et al., 2016; Miller et al., 2016; Say et al., 2014; Tunçalp et al., 2017). Additionally, shocks originating from within and beyond the healthcare system—such as unfavorable environments, adverse socioeconomic and political climates, gender-based inequities, and the absence of sustainable social justice frameworks—lead to further abuse. Therefore, many of the world's most vulnerable women are not provided these services, and even where available they cannot access the high-quality services essential for positive health outcomes for all women and their newborns (Ceschia & Horton, 2016; da Silva, 2019; Kinney et al., 2016; Knaul et al., 2016; Koblinsky et al., 2016; Miller et al., 2016).

In contrast, most HICs boast the availability of high-quality maternal and newborn health services, with high levels of antenatal care (ANC) attendance, facility-based births, safe abortion care services, and high-quality family planning services. Additionally, the presence of robust health systems and the availability of innovative health system financing means most women and girls can access life-saving maternal health services that lead to better maternal and neonatal health outcomes (Graham et al., 2016; Shaw et al., 2016). Yet some question whether gains in healthcare provision are threatened by a culture of hyper-medicalization (that is, providing “too much too soon,” including the routine use of interventions with no proven health benefit for women or even interventions that are potentially harmful or unnecessarily costly) (Miller et al., 2016).

As a result, it has been observed that low maternal mortality HICs and some LMICs with improving rates of maternal mortality report abnormally high levels of specific obstetric interventions, including the induction and augmentation of labor, unnecessary ultrasound examinations (Miller et al., 2016), and what has been termed the *Caesarean section epidemic* (or an excess in elective C-section births) (Boerma et al., 2018). The increased cost of care associated with these interventions can lead to disparities in access, often disadvantaging marginalized populations such as women of color, ethnic minorities and indigenous populations, women and girls with disabilities, refugees and internally displaced persons, and adolescent girls (Carniato, 2021; Huang et al., 2018; Kasper et al., 2022; Kazembe et al., 2022; Leonard et al., 2019; Shaw et al., 2016).

Hence, it is important to approach the analysis of maternal healthcare through a rights-based equitable perspective. This includes examining the quantity, quality, and efficiency of these services worldwide, even in HICs that manifest better maternal and newborn health outcomes compared to their LMIC counterparts.

Decision-Making, Agency, Autonomy, and Empowerment in Maternal Health

Maternal health is related to countless decisions for individual women, such as when to become sexually active, whether to use contraception, when to become pregnant, whether to carry a baby to term, where to give birth, and when to seek healthcare. This relationship between women's decision-making power and their sexual and reproductive (including maternal) health outcomes was first affirmed as a policy statement at ICPD 1994 (Population Council, 1995), which coincided with the affirmation of women's inalienable right to access comprehensive sexual and reproductive health (SRH) services of their choice. More than 50,000 participants and activists and 189 governments defined a platform for action in an unprecedented move toward achieving gender equality and empowering women worldwide (UN Women, 2015). Such commitments have since been reaffirmed at ICPD 25 by the International Planned Parenthood Federation (IPPF, 2019).

However, even 30 years after ICPD 1994, huge implementation gaps exist with regard to commitments made by governments. Some of the most affected areas include adolescent access to SRH information, education, and services, as well as equity gaps that continue to exclude the poorest and least-educated rural women and girls (Hardee & Jordan, 2021; Lusti-Narasimhan et al., 2014; Manguro & Temmerman, 2022; Pratley, 2016). Yet, United Nations data suggest that between 2007 and 2021, 57% of women aged 15 to 19 who were married or in a union reported making their own informed decisions regarding sexual relations, contraceptive use, and reproductive healthcare (Sustainable Development Goals [SDG], 2022; UN-DESA, 2015). This suggests that 43% of women globally did not make their own decisions about such vital issues in maternal health.

Studies in a number of low-resource settings indicate that women often are not able to make decisions about seeking healthcare for themselves or their infants (Gupta et al., 2015; Osamor & Grady, 2016). One study based in South Asia found that of ever-married women of reproductive age, only 13.4% in Nepal, 28.1% in India, and 17.6% in Bangladesh made decisions alone about their own healthcare, and the majority of such decisions were made without the woman's participation at all in Bangladesh (54.3%) and Nepal (72.7%) (Senarath & Gunawardena, 2009).

Women being able to make their own decisions about their health can be referred to as *women's agency*, *women's autonomy*, or *women's empowerment*—all constructs that were initially developed in the West and have not necessarily addressed how such factors may differ in cultures defined by a more collectivist than independent orientation (Iyengar & DeVoe, 2003; Markus & Kitayama, 2003). For example, a qualitative study by Moyer et al. (2012) highlighted the role that grandmothers play in care-seeking decisions for mothers and newborns in Ghana, and the importance of considering the influence of collectivist community structures when designing interventions (Gupta et al., 2015).

Nonetheless, the concepts of agency, autonomy, and empowerment have been studied in relation to maternal health in both HICs and LMICs (Nieuwenhuijze & Leahy-Warren, 2019). Some researchers view these concepts as multidimensional, encompassing psychological, social, economic, and legal aspects (Pratley, 2016), and as enabling material (income, education) and

social resources (later sexual debut, marriage, and childbirth) that allow a woman to make choices about her own life (Kabeer, 1999). Still others rely on available data from Demographic Health Surveys (including questions about household decision-making, attitudes toward intimate partner violence, freedom of movement, and ownership of assets) to define empowerment through the development of multi-item scales (Asaolu et al., 2018; Ewerling et al., 2017; Miedema et al., 2018).

Even without universal definitions or measurement strategies, the concepts of agency, autonomy, and empowerment have been linked to better maternal and newborn outcomes. In a systematic review of the evidence published in 2016, Pratley found statistically significant associations between women's empowerment and maternal and child health outcomes, including ANC utilization, skilled attendance at birth, contraceptive use, child mortality, children's vaccination status, child nutritional status, and exposure to violence (Pratley, 2016). Although the strength of the associations varied, the studies included in the review supported the hypothesis that women's empowerment is significantly and positively associated with maternal and child health outcomes (Pratley, 2016). Therefore, in addressing the different facets of maternal health and well-being, it is important to bear in mind the prevailing impact of women's autonomy, agency, and empowerment.

Maternal Health Epidemiological Burden

Approximately 210 million women and girls are estimated to become pregnant each year, resulting in an expected 140 million births (Graham et al., 2016). Based on World Health Organization (WHO) estimates, about 295,000 women and girls died in 2017 alone as a result of pregnancy/childbirth (WHO, 2019). While temporal trends in global the maternal mortality ratio (MMR) give an optimistic perspective to the improvement of maternal health, regional variations highlight inequities in access to high-quality care for those living in LMICs. Eighty-six percent of maternal deaths occur in sub-Saharan Africa and South Asia ("Cairo + 25," 2019). Women in sub-Saharan Africa have a 1:36 lifetime risk of maternal death, compared to 1:4,900 among women in HICs, due in large part to barriers to obtaining timely, high-quality care (Alkema et al., 2016; Graham et al., 2016).

In contrast, as rates of maternal mortality remain low in many HICs, focus has shifted to reducing maternal morbidity. The WHO defines *maternal morbidity* as any health condition attributed to and/or aggravated by pregnancy and childbirth that has negative outcomes to the woman's well-being. In the United States, which has one of the highest rates of maternal mortality among HICs, rates of severe maternal morbidity (SMM) are monitored to measure the short- and long-term health consequences of pregnancy and delivery. While SMM has not been adopted as an international indicator of maternal health, monitoring for SMM—for example, pregnancy-related heart conditions, eclampsia, and end-organ damage—in LMICs would demonstrate maternal health inequities to be greater than perceived (American College of Obstetrics and Gynecology et al., 2016).

At the same time, demographic, epidemiological, environmental, and socioeconomic transitions are responsible for the increasing contribution of noncommunicable diseases to mortality and morbidity among mothers (Graham et al., 2016; Say et al., 2014). Specifically, pregnancy has been described as a cardiovascular stress test because the onset of specific pregnancy complications could suitably predict future cardiovascular events in women and girls (Cusimano et al., 2014; Sharma et al., 2021; Smith, 2015; Smith et al., 2013). In addition to cardiovascular pathologies, hyperglycemia in pregnancy, either due to preexisting diabetes or resulting from gestational diabetes, has been linked to adverse maternal, fetal, and neonatal adverse outcomes, and is also associated with epigenetic mechanisms that ultimately place the exposed newborn at risk of diabetes and other cardiovascular risk factors in adult life (Hare et al., 2020; Hjort et al., 2019; Mackin et al., 2018, 2019). Furthermore, unhealthy weight gain/retention in pregnancy and the puerperium is a risk factor for obesity, which in turn exposes the woman to a myriad of future cardiovascular events such as myocardial infarction, hypertensive heart disease, diabetes, and stroke (Cusimano et al., 2014; Smith et al., 2013).

In addition to the clinical impact of these noncommunicable diseases, the costs linked to prevention and screening impose additional strain on health systems worldwide, revealing equity-linked gaps as the women in the world's poorest regions and women and adolescent girls in the lowest wealth quintiles are least likely to have access to care despite being disproportionately burdened by these conditions (Hare et al., 2020; Smith et al., 2013). Therefore, pregnancy-related care could provide a strategic opportunity for screening and to raise awareness about chronic noncommunicable diseases such as hypertension and diabetes. It could also initiate interventions aimed at preventing adverse disease-linked events and could, ultimately, improve women's overall health outcomes.

Another area that is increasingly attracting research and policy attention globally is maternal mental health. Evidence indicates that maternal mental health is both essential for a woman's well-being as well as for the health and well-being of the fetus, the newborn, the child, the adolescent, the family, and her community in the immediate as well as long term (Alderdice & Newham, 2016; Bennett et al., 2016; Howard, Molyneaux, et al., 2014; Howard, Piot, & Stein, 2014; Jones et al., 2014; Stein et al., 2014). Perinatal mood and anxiety disorders (PMADs) constitute the most burdensome mental and neurological condition affecting women of reproductive age, yet women's mental health in and out of pregnancy and childbirth is yet to receive as much attention as their physical health worldwide, and especially in LMICs (Atif et al., 2015). These disorders on this spectrum, for which depression and anxiety are the most common, are a frequent complication of pregnancy and are responsible for adverse maternal outcomes (Camacho & Shields, 2018; Moore Simas et al., 2019). Ultimately, PMADs and perinatal suicidality contribute to as much as 20% of postpartum deaths in HICs, while the data in LMICs remains hard to obtain (Chin et al., 2022; Slomian et al., 2019).

Beyond the mother's health, maternal mental disorders have been variably associated with poor fetal outcomes (low birth weight, prematurity, intrauterine growth restriction), infant death, as well as psychological, emotional, and developmental disturbances in children, and even an increased risk of severe depression in adolescents. However, it is difficult to measure clearly if these complications are due to the disease itself or due to the use of certain medications whose

safety during pregnancy has not been established. Above all, maternal mental health has untold and lasting impact on the community. However, gaps exist with respect to attributing these disorders to pregnancy, particularly during the postpartum period when symptoms could occur well beyond the puerperium (Chin et al., 2022; Slomian et al., 2019).

The gaps in identifying and addressing PMADs are especially wide in LMICs. WHO data suggests that only 10% of LMICs reported maternal mental health findings compared to 90% of HICs, despite findings indicating that perinatal mental health diseases may be more prevalent among women in LMICs (Alderdice & Newham, 2016; Atif et al., 2015; Bennett et al., 2016). There are also significant management gaps in LMICs, especially for severe forms of mental disorders (Howard, Molyneaux, et al., 2014; Howard, Piot, & Stein, 2014; Jones et al., 2014). Management gaps are especially problematic because the coexistence of maternal mental health disorders with infectious comorbidities such as HIV or TB are more common in LMICs, putting both mother and babies at elevated risk of poor outcomes (Howard, Piot, & Stein, 2014).

Maternal Health as a Determinant of Women, Newborn, and Children's Health Outcomes

Women's rights champion Mahmoud Fathallah said; "Women are not dying of diseases we cannot treat; they are dying because societies have yet to make the decision that their lives are worth saving" (Bergström, 2016, p. 2). In a world where the most vulnerable women and girls live in resource-limited communities that are dependent on aid to provide maternal health services (GrantWatch, 2021; Pitt et al., 2021), and in a time when COVID-19 has significantly distorted the progress made in the field (Chmielewska et al., 2021; Kotlar et al., 2021; Pitt et al., 2021), why should governments and international donors prioritize the delivery of high-quality maternal health services?

The answer lies beyond the need to fulfill a moral obligation to one another as humans. Investing in health systems to provide high-quality maternal health services offers a high return on investment, with significant intergenerational impacts (Chmielewska et al., 2021). Mother and newborn are inextricably linked, and investing in the provision of high-quality maternal health services not only addresses maternal health targets but can also avert preventable neonatal and child deaths, as well as help ensure optimal health and developmental outcomes for children (Kinney et al., 2016; Lassi et al., 2013).

Prioritizing maternal health services is an important component of ensuring high-quality care, but initiatives that incorporate an integrated continuum-of-care strategy are ideal to avoid competition across maternal, newborn, and child health (MNCH) efforts for increasingly scarce resources. While making the business case for high-quality maternal health services, governments and their partners must insist on strengthening health systems as the surest path to any sustainable wins for mothers, newborns, and children. There is a need for global and national accountability at all levels to ensure the kind of progress that is respectful of the needs and wishes of women and girls worldwide.

Maternal Health: What Are the Issues?

Analyzing maternal care from a human rights lens demands the international community incorporate issues related to maternal health that expand beyond pregnancy. At its root, women's agency, and the ability for women to define their own health choices and actions, is the foundation of providing women with dignifying healthcare. Consequently, family planning enhances the foundation of women's agency by providing women with the requisite options to exercise their reproductive rights and make their own reproductive health choices. Next, whenever and wherever women and families desire and decide to achieve pregnancy, the provision of preconception care, including infertility services, is essential as it can address preventive measures, lifestyle factors, and specific needs for a successful pregnancy journey while also serving as an intersection between maternal and paternal health both essential to fetal and newborn outcomes. Thus, once pregnant, adequate antepartum, intrapartum, and postpartum care are imperatives toward reducing maternal, fetal, and newborn morbidity and mortality.

With this in mind, maternal health services worldwide often fall short of providing evidence-based, safe, effective, timely, and efficient interventions. Additionally, these services are not always rights based or person centered. Often communication between providers and patients is suboptimal, precluding women from feeling as though they are a part of the decision-making process in issues relating to their own bodies. Ultimately, healthcare is not always provided in ways that upholds the dignity of women and girls or ways that safeguard their social, cultural, and emotional well-being (Downe et al., 2018; Kinney et al., 2016; Tunçalp et al., 2015; WHO, 2006).

Planning the Pregnancy

The London Family Planning Summit in 2012 marked a resurgence in family planning efforts with the establishment of the "120 by 20" goal. This goal, set by leaders of 150 countries, international agencies, civil society organizations, foundations, the research and development community, and the private sector, aimed to increase the number of modern contraception users worldwide by 120 million by 2020, while maintaining the 2012 base of 260 million users of modern contraception (Brown et al., 2014; Stover & Sonneveldt, 2017). Beyond addressing contraceptive numbers, this summit ushered in the FP2020 and FP2030 global initiatives, which served to mobilize resources and ensure global consensus on the need for high-quality voluntary family planning services, against the backdrop of meeting girls' and women's unmet needs as well as their right to practice contraception (Brown et al., 2014; Cahill et al., 2018; Stover & Sonneveldt, 2017).

Yet boosting contraceptive use is not necessarily a straightforward endeavor. Contraceptive uptake is linked to many factors affecting women, their partners, and their communities, including knowledge, attitudes, access, cost, cultural expectations, and social norms. Women's empowerment appears to be tightly linked with contraceptive uptake (Prata et al., 2017). Data

from nearly 500,000 women of reproductive age in 32 countries in sub-Saharan Africa showed a consistent relationship between various components of women's empowerment and contraceptive use after adjusting for demographic and socioeconomic factors (Yaya et al., 2018).

Consequently, even with substantial worldwide investments, the number of women who had access to modern contraception increased by an additional 87 million, below the FP2020 target of 120 million new users. Concurrently, 162.9 million women had an unmet need for family planning, with more than half of these women residing in either sub-Saharan Africa or South Asia. Young people aged 15 to 24 made up more than a quarter of those with unmet family planning (FP) needs (Family Planning, 2020, 2022; Pradhan, 2022). Furthermore, the lack of a broad contraceptive mix, the dominance of female sterilization associated with limited healthcare autonomy, and the selective exclusion of adolescents from accessing comprehensive sexual and reproductive health services are key gaps worldwide (Brown et al., 2014; Hardee & Jordan, 2021; Pradhan, 2022).

Preconception Care

Though evidence remains scarce, existing sources indicate that preconception health is a key determinant of pregnancy success and next-generation health (Barker et al., 2018; Fleming et al., 2018). Yet preconception care remains an underused service worldwide. Even when used, the focus is primarily on micronutrient supplementation with folic acid and iron prior to conception (De-Regil et al., 2010; Mastroiacovo & Leoncini, 2011). Very little attention is given within the maternal healthcare continuum to sensitive periods (adolescence, no pregnancy desire) and/or the cumulative effects of unhealthy exposures (poor diet, alcohol, smoking, obesity, etc.) to maternal health and well-being (Fleming et al., 2018; Stephenson et al., 2018).

Furthermore, pregnancy and childbirth are key contact periods between women, girls, and the health system. Conversely, the configuration of healthcare services often fails systematically to provide the distinct requirements of adolescents, adults not intending to get pregnant, and adults who want to achieve pregnancy, while paternal health as a determinant of pregnancy outcomes is almost completely excluded from the discussion (De-Regil et al., 2010; Fleming et al., 2018; Stephenson et al., 2018).

Antenatal Care

Irrespective of the model, historically the uptake of antenatal care (ANC) in high-income countries (HICs) has been as high as 98%, aided by properly functioning health systems, significant socioeconomic welfare, and innovative advances in clinical care (Zanconato et al., 2006). Conversely, although low-income countries have registered remarkable progress in women's uptake of ANC services following MDG and now SDG-influenced policy efforts, overall uptake remains low. Between 2011 and 2016, only 52% and 46% of pregnant women from sub-Saharan Africa and South Asia, respectively, indicated that they had received the recommended number of four ANC visits (Obse & Ataguba, 2021).

In addition to regional and inter-country disparities, an analysis of ANC trends from low-uptake regions demonstrates huge inequities but also huge quality deficits. These inequities have been reported with the uptake of a minimum of four ANC visits, for which women and girls in the poorest quintiles are selectively disfavored, as are those in rural settings, adolescents, single women, and those with little or no education (Abekah-Nkrumah, 2019; Bain et al., 2022; Obse & Ataguba, 2021).

Another gap that has affected ANC within the continuum of care has been the historical lack of consensus within the maternal health community regarding the relevance of ANC, as well as how best to deliver ANC interventions, and the quality and interpretation of the evidence that has informed different interventions and models. So, whereas some authors intimate that ANC has no true impact on maternal health outcomes, others suggest that ANC could have a dose-response effect on maternal health outcomes, while some voices are convinced that high-quality, evidence-based ANC interventions could significantly influence maternal health outcomes (Baldo, 2001; Carroli, Rooney, & Villar, 2001; Hall et al., 1980; Tunçalp et al., 2017).

Specifically, in 1978, the World Health Organization (WHO) validated a managerial strategy that aimed to provide ANC to women by evaluating them for individual risk (risk approach), but eventually backtracked after it became increasingly evident that most obstetric emergencies occurred in otherwise low-risk pregnant women (Zanconato et al., 2006). As maternal deaths became more visible in low-income settings, the Safe Motherhood Initiative was introduced, and this favored investments in functional referral systems and sought to redistribute the scarce resources available for healthcare (Potts, 1990; Weil & Fernandez, 1999). The aim was to encourage the uptake of facility-based, skilled care for women in pregnancy and childbirth, while ensuring that women in need could access emergency obstetric and neonatal care services.

With time, sociocultural, economic, and human considerations raised questions as to the adequacy of traditional ANC models in low-income settings. Specifically, questions were raised regarding the efficacy of the remarkable number of hospital visits and the requirement that women undertake a plethora of clinical and paraclinical tests. This led to the WHO ANC trial and to the proposed goal-oriented ANC approach. The approach was credited with delivering effective and appropriate screening, preventive measures, and treatment interventions in a timely way without increasing the costs of care or the rates of adverse maternal, fetal, and newborn outcomes as compared to the traditional eight-visit model (Carroli, Villar, et al., 2001; Villar et al., 2001).

Eventually, the WHO undertook to reanalyze the data from the initial trial after secondary evidence from within the maternal health community linked the reduced visit model to a 15% increase in perinatal mortality (Hofmeyr & Hodnett, 2013; Vogel et al., 2013). In addition to confirming this risk of four extra perinatal deaths per 1,000 births in a typical resource-limited setting, the WHO's guideline development group further indicated that the previous model was not fully considerate of the express wishes of women who had voiced dissatisfaction with the fewer visits (Downe et al., 2016) offered via the basic ANC model. This constituted the basis for the WHO's new positive pregnancy experience ANC guidelines in 2016, which sought to ensure the integrated delivery of key evidence-informed interventions while proposing ways in which

countries can rethink, redesign, and thus strengthen their health systems to provide quality ANC in ways that are respectful of women's rights and needs (de Masi et al., 2017; Tunçalp et al., 2017; WHO, 2016).

Despite its innovative strategies and rigorous methods, the uptake of some of the recommendations may be subjected to future scrutiny owing to a lack of consensus on the strength of evidence or the validity of the recommendations across different settings worldwide (Weeks & Temmerman, 2016), while some voices from LMICs seem to indicate that it might not be a good fit for resource-limited settings (Mchenga et al., 2019; Obse & Ataguba, 2021).

Intrapartum Care

In the fight against preventable maternal and newborn deaths, an essential strategy has been to focus on and ensure the provision of skilled birth services. This approach has led to a remarkable increase in the coverage of facility-based deliveries worldwide (Campbell et al., 2006; Montagu et al., 2017; WHO, 2018a). The WHO reported that 84% of all births worldwide were assisted by a skilled health worker between 2015 and 2021, compared to 64% between 2001 and 2007 (WHO, 2022c).

Unfortunately, increases in the uptake of skilled deliveries, and the known positive impact of the availability of skilled birth attendants during childbirth, have not translated into better maternal and newborn morbidity and mortality outcomes in the world's high-mortality burden regions. The stalling progress in maternal and newborn outcomes in sub-Saharan Africa and Southeast Asia has been attributed to various health system access and demand gaps, but also to the questionable quality of intrapartum care services where and when available (Bhutta et al., 2014; Montagu et al., 2017; Tunçalp et al., 2015).

Regarding access, the three-delays model is a framework utilized in examining barriers in maternal health. In Thaddeus's "Too far to walk maternal mortality in context," (a) delay to seek appropriate care, (b) delay in reached obstetric facility, and (c) delay in adequate care at facility are outlined as contributing factors in the limitations to appropriate maternal care. While initially designed to better understand barriers to care during obstetric emergencies, the three-delays framework can be used to identify the personal, socioeconomic, and political influences that result in challenges at the household, community, and health system levels in LMICs (Actis Danna et al., 2020).

Regarding quality of care, the WHO defines *quality* as "the extent to which health care services provided to individuals and patient populations improve desired health outcomes" (WHO, 2006, p. 9). By referring to desired health outcomes, the complex and multidimensional concept of "quality of care" inadvertently introduces the even grayer, and nonbiological, concept of women's experiences, which are, however, inextricably linked to the whole concept of quality care.

The importance of labor and delivery as the ultimate physiological pregnancy outcome for the mother–newborn duad has declined over time with the increased use of planned caesarean deliveries in many settings worldwide (Buchmann et al., 2016). Yet, the birthing process remains unavoidably important due to the ever-present possibility of adverse maternal, fetal, or newborn outcomes. In line with this, the WHO reported that between 1990 and 2015, an estimated one in every three maternal deaths occurred during labor and delivery (Buchmann et al., 2016; Kassebaum et al., 2014). Other sources reported much higher figures, estimating that about 70% of maternal deaths are intrapartum—mostly due to hemorrhage, preeclampsia/eclampsia, and sepsis (Say et al., 2014).

Unfortunately, *normality* within the context of intrapartum care is yet to be standardized worldwide, while the historical understanding of *normal labor* based on trials carried out in the previous century might fall short in the light of rights-informed and person-centered evidences (Oladapo, Diaz, et al., 2018). Thus, intrapartum care models vary significantly on issues regarding the place of birth, the frontline caregiver, and care patterns and philosophies, including access to the woman's choice of support during labor and delivery. The availability of resources also seems to be a major determinant of the intrapartum care model across different settings.

While in most resource-rich settings intrapartum care is led by midwives, family physicians, or obstetricians, with some settings practicing shared models of care, the shortage of skilled maternity workers in sub-Saharan Africa meant health systems had to depend on traditional birth attendants for many decades (Oladapo, Tunçalp, et al., 2018; Renfrew et al., 2014). Existing evidence seems to favor midwife-led care over other models of care, even though gaps exist when it comes to authoritatively indicating which care model offers the best maternal and newborn health outcomes and how best to ensure the effective uptake and implementation of such a model across different resource settings (Oladapo, Tunçalp, et al., 2018). Nonetheless, the availability of basic and comprehensive emergency obstetric services seems to be more important for maternal–fetal outcomes than skilled birth care on its own, regardless of who the skilled attendant might be (A. C. C. Lee et al., 2011).

One major challenge in intrapartum care, predominantly in middle- and high-income countries, is related to what some call the *overmedicalization* of the labor and delivery process. Overmedicalization involves treating all pregnancies as medical events warranting intervention, which aligns with the significant rises seen globally in labor induction, oxytocin augmentation, caesarean sections, operative vaginal birth, or episiotomies to initiate or speed up labor and delivery. This ultimately imposes significant economic costs on both the healthcare system and the individual woman, and excess interventions can undermine the goal of rights-based, person-centered care (Gibbons et al., 2012; Villar et al., 2006). In addition to potentially unnecessary clinical interventions, women may be subjected to such things as perineal shaving, enemas, amniotomy, intravenous fluids, antispasmodics, and so on, although there is no evidence of improved maternal and newborn outcomes (Oladapo, Tunçalp, et al., 2018; Renfrew et al., 2014; WHO, 2018c).

Caesarean deliveries are life-saving procedures that are essential in the management of specific maternal, fetal, and neonatal complications that might occur during pregnancy or arise during labor and delivery (Betrán et al., 2016; Villar et al., 2006). However, this procedure is often undergone for nonurgent, nonclinical indications, such as for convenience of the provider or preference of the patient, particularly in contexts experiencing overmedicalization of the birthing process (Boerma et al., 2018; Miller et al., 2016). While C-sections may indeed be lifesaving, the procedure is also associated with immediate, medium-, and long-term maternal and perinatal risks. Nonetheless, caesarean deliveries have increased globally and account for approximately one out of every four births worldwide, well above the optimal 10% to 15%. Increases are most notable in Latin America and the Caribbean subregions, where caesarean levels as high as 44% have been reported, even though rates as high as four times the maximum threshold recommended by the WHO have been reported in Egypt (Wahdan et al., 2022). This “caesarean section epidemic” has been associated with socioeconomic development, greater female enrollment in secondary education, physician densities, low fertility, and greater urbanization. Excessive use of caesarean sections for deliveries not only significantly increases the cost of care but it also raises questions about both the rights of the patient and the patient’s agency (Kinney et al., 2016; Tunçalp et al., 2015; Villar et al., 2006; WHO, 2006). In the case of an elective C-section, should a patient’s preference outweigh medical necessity?

While caesarean delivery rates are increasing globally, it is important to note that huge access gaps still exist in the world’s high-mortality regions, where fewer than 10% of deliveries occur by caesarean delivery. In regions such as sub-Saharan Africa, the shortage of basic and comprehensive emergency obstetric and newborn care (EmONC) services, the shortage of skilled personnel, high fertility rates, low female secondary school enrollment, and a predominantly rural population have been associated with low rates of C-sections both between countries and among different populations within the same country (Betrán et al., 2016; Gibbons et al., 2012; Kinney et al., 2016; Villar et al., 2006). In these settings, the absence of essential caesarean delivery services reveals gaps in equity and the quality of care.

Postnatal (Postpartum) Care

The WHO defines the *postnatal period* (also *postpartum* or *puerperium*) as the period right after childbirth and spanning the first six weeks following delivery (WHO, 2022c). Although the timeframe of six weeks seems to have some universal acceptability, it is important to note that the duration of the postnatal period is subject to cultural variability. Therefore, unlike the WHO, the National Institute for Health and Care Excellence recommends an eight-week postnatal window following childbirth (National Institute for Health and Care Excellence [NICE], 2021). This is especially important given that some pregnancy complications such as perinatal mood and anxiety disorders and other mental health conditions could persist well beyond the first year of birth, at which time the woman might no longer be in contact with the health system (Howard, Piot, & Stein, 2014; Overbeck et al., 2022; Slomian et al., 2019).

Although there is consensus regarding the importance of postnatal care (PNC), it remains the most neglected component of maternal healthcare services (Sacks & Langlois, 2016). “Countdown to 2030” data reveals that within the maternal, newborn, and child health continuum, PNC services have the lowest median national coverage worldwide (UNICEF, WHO, 2017). The percentages for coverage within two days after birth for women (71%) and newborns (64%) are a long way from 2025 targets, which aim to ensure that every woman and newborn receives early routine PNC within two days (Requejo et al., 2020; UNICEF, 2020).

Furthermore, while PNC is nearly universal in many HICs, there is significant inter-country and intra-country variation in the uptake of PNC services. This holds especially true in LMICs, which present specific barriers related to socioeconomic, cultural, and health system factors (Langlois et al., 2015). Reports indicate that the median coverage of PNC was 62% for mothers in the 51 countries in the world with the highest maternal deaths, while the median coverage of PNC for babies was 36% for the 48 countries in the world where child death rates are the highest (UNICEF, WHO, 2017).

The low uptake of PNC services in LMICs has been linked to significant socioeconomic and geographic inequities in access to services (Langlois et al., 2015; Singh et al., 2012), including availability of services, human and financial resources, sociocultural norms and practices, women’s experience of care, and health literacy (Dhaher et al., 2008; Kinney et al., 2016; Munabi-Babigumira et al., 2021; Sacks et al., 2022; Tunçalp et al., 2015). But simply increasing the number of mothers and newborns who obtain PNC is unlikely to have a significant impact absent considerations of quality. Early discharge, poor monitoring, deficient patient counseling, and the fragmentation of services following discharge represent some of the quality deficiencies that ultimately affect the impact of postnatal care services (Bick et al., 2020; Munabi-Babigumira et al., 2021). In addition, as with most mother-newborn-child health interventions, there is little global consensus as to what constitutes “normal practice,” hence the variations in global practices. At the same time, variability across health systems prevents the consistent implementation of evidence-based practices, even when strong evidence exists. Nonetheless, the WHO has been a key player in the formulation of recommendations for PNC, specifically in LMICs.

Global consensus indicates that high-quality PNC is a critical focus for research, policy, and practice (Benova et al., 2019; Pitt et al., 2021; Sacks & Langlois, 2016; Temmerman, 2016). However, while available evidence describes the facilitators and barriers to the uptake of PNC (Benova et al., 2019; Munabi-Babigumira et al., 2021; Sacks et al., 2022), while very few studies examine the perceived and actual quality of care or the disrespect and abuse during care provision and how these ultimately impact the uptake of services (Kujawski et al., 2015; Vogel et al., 2016). Given the recommendation for an integrative approach to postnatal consultation for the mother-baby dyad, significant evidence gaps exist in the discrepancies between maternal and newborn postnatal coverage (Amouzou et al., 2020).

Maternal Health: The Opportunities

The Sustainable Development Goals (SDGs) aim to end all preventable maternal and newborn deaths by 2030 (Brizuela & Tunçalp, 2017; Temmerman, 2016; UN-DESA, 2015). To achieve this, the effective uptake of high-quality family planning, preconception, antenatal care (ANC), assisted and facility-based skilled deliveries, and postnatal care (PNC) will be essential (Bain et al., 2022; Schisterman et al., 2014; Shaw et al., 2016; Stephenson et al., 2018). Data suggests that ensuring the widespread availability of high-quality antenatal, delivery, and postnatal services could prevent 71% of maternal and newborn deaths by 2025, albeit at a cost of about \$2,000 per maternal or newborn life saved (Bhutta et al., 2014).

The SDG agenda has ushered in a significant shift from just ensuring survival to an increasing focus on notions of *thrive* and *transform* (United Nations, 2015). This shift opens up a realm of opportunities to women worldwide, albeit to varying extents. In line with this renewed vision, the World Health Organization (WHO) is now referencing a positive experience, defined as follows:

Maintaining physical and sociocultural normality, maintaining a healthy pregnancy for mother and baby (including preventing or treating risks, illness and death), having an effective transition to positive labour and birth, and achieving positive motherhood (including maternal self-esteem, competence and autonomy).

(Tunçalp et al., 2017; WHO, 2018c, 2022c, p. 860)

A detailed appraisal of maternal mortality trends reveals glaring worldwide disparities. While worldwide improvements give hope for the future, they also reveal a dynamic process best known as the *obstetric transition*, which provides guidance on what different countries might need to prioritize in the immediate term to end these deaths (Souza et al., 2014). Thus, for countries in sub-Saharan Africa and South Asia where fertility remains high, as well as maternal mortality mostly due to direct causes, the focus must be on improving access by ensuring the availability of basic infrastructure and working toward improving female literacy, women's agency and empowerment, access to high-quality contraceptive services, and prevention of unsafe abortion, as well as investing in health system strengthening and especially capacity building for skilled workers, with an insistence on the quality of care (Kinney et al., 2016; Koblinsky et al., 2016; Souza et al., 2014; WHO, 2006). Conversely for some middle- and high-income countries and health systems, the focus must be on improving the quality of care, investing in secondary and tertiary prevention, avoiding health system delays, and limiting the overmedicalization of care (Kinney et al., 2016; Koblinsky et al., 2016; Sharma et al., 2021; Souza et al., 2014; WHO, 2006).

Reportable measures to monitor the quality of maternal health are being designed to improve maternal health for women worldwide. The ongoing process of monitoring trends in maternal outcomes provides insight to health disparities among various conditions impacting women across the globe. Furthermore, this process aims to provide key information that informs health policies and interventions with the aim of more equitable access to maternal care for all. Health organizations worldwide continue to work on providing standardized measures of quality and access to inform the global community on areas for improvement within maternal health.

Maternal Agency, Autonomy, and Empowerment

One overarching opportunity in the maternal health space is the development, implementation, and evaluation of interventions aimed at improving women's sense of agency, autonomy, and empowerment. While the literature is replete with data suggesting a strong relationship between higher autonomy and better health outcomes, very little is actually known about the best ways to boost autonomy and how that might vary by culture, population, or setting.

Some researchers have explored financial mechanisms to boost empowerment, including cash transfers, savings and loan groups, and other microfinance mechanisms (H. Lee et al., 2021), but few have documented changes in agency, autonomy, or empowerment over time. One notable exception is the 2019–2021 “Men Engagement in Women Empowerment for Improving Maternal Mental Health Through Cash Transfer and Life Skills Building” trial, or the MEWE trial, in Pakistan (Bhutta et al., 2019). This trial, results pending, sought to explore the effects of a skill-building and cash-transfer program on empowering women and couples. Other researchers have focused on stigma-reduction frameworks to boost empowerment among vulnerable groups, such as pregnant women with HIV (Poku et al., 2020), and it is imperative for future research to focus on how best to improve maternal agency, autonomy, empowerment, and, in turn, maternal health.

Healthcare workers' inability to integrate notions of women's agency, rights, and autonomy has been cited among access-related barriers. To address this, there are increasing calls to either systematically integrate or improve sexual and reproductive health and rights content in medical schools' curricula (Endler et al., 2022; Pugh, 2019; WHO, United Nations Children's Fund, World Bank Group, 2018). Although evidence remains rare in the area, some authors suggest that it would reduce SRH violations within the health system, improve access and uptake, and ultimately lead to better SRH outcomes, in particular, as well as better health outcomes in general for women worldwide (Endler et al., 2022; Starrs et al., 2018).

Preconception Care

Available clinical, biological, social, and policy field evidence shows that preconception health is vital for pregnancy success as well as next-generation health (Barker et al., 2018; Fleming et al., 2018; Stephenson et al., 2018). Therefore, the preconception window could provide vital opportunities to improve both paternal and maternal indicators necessary to the well-being of a pregnant woman and her baby, but also the long-term health of her baby once born.

When identifying intervention windows within the preconception period, healthcare systems must consider the biological, individual, and public health perspectives—with each distinctly and significantly impacting the available time for any sort of intervention—but above all must consider the intention at the individual level to conceive or not. Whatever the situation, the preconception provides a window for education on and the provision of healthy nutritional options (including micronutrient supplementation). It also allows for communication regarding behavior change, specifically addressing risk factors such as obesity, sedentary lifestyle, and substance abuse. In addition, it enables screening, detection, and management of potentially

morbid conditions that could threaten the well-being of the pregnant woman or the newborn (Barker et al., 2018; Lawn et al., 2016; Sahariah et al., 2016; Smith et al., 2013; Stephenson et al., 2018).

The preconception window presents an opportunity to educate individuals about sexually transmissible infections (STIs), screen for these infections, and provide appropriate care—and thus ensure both the better health and well-being for mothers and their partners and the prevention of harmful fetal exposures if pregnancy is desired or arises (Lassi, Dean, et al., 2014; Lassi, Imam, et al., 2014; Richardson & Shannon, 2012). It is worth noting that persons who are not pregnant might have contact with maternal health services. It therefore becomes the explicit responsibility of the entire health system to ensure that, at every contact, women and girls are empowered to make decisions that best align with their pregnancy intentions and that ensure healthy lives for them and their babies if they choose to get pregnant.

Furthermore, evidence suggests that when the broader society recognizes the importance of preconception care, it has the potential to enhance preconception health and promote positive health-seeking behaviors within the population. Such societal involvement can also garner support from political policymakers and thus contribute to achieving sustainable results (Barker et al., 2018; Sahariah et al., 2016). Some countries have attributed improved preconception nutrition for women as a contributing factor to the eradication, between 1994 and 2006, of undernutrition and wasting among children younger than 5 years old, the eradication of maternal diabetes (Fleming et al., 2018; Hare et al., 2020; Sahariah et al., 2016), and the screening for and treating of STIs (Lassi, Dean, et al., 2014; Lassi, Imam, et al., 2014).

Antenatal Care

ANC consists of a strategic set of interventions within the maternal health continuum of care for women and children's health in general. ANC represents the entry point for most women and their children into the health system, and with regard to maternal health, it represents an opportunity for health education, health promotion, and disease screening and prevention in all settings worldwide (Mamba et al., 2017; Tunçalp et al., 2017). Furthermore, potentially comorbid conditions like malaria in pregnancy, hypertensive disorders of pregnancy (including preeclampsia and eclampsia), and anemia can be detected early and managed appropriately. Therefore, specific high-quality and evidence-based ANC interventions could significantly improve maternal health outcomes.

The well-being and survival of the mother-baby duad is indissociable, and a positive pregnancy experience requires a positive outcome for both mothers and their newborns. More than half (55%) of stillbirths occur in the antepartum period, yet such numbers are often not captured or included in MNCH efforts. Thus stillbirths tend to be underreported worldwide, including in high-burden LMICs (Lawn et al., 2011, 2016; United Nations, 2015). Because the majority of stillbirths occur in the antepartum period, and most neonatal deaths are linked to preventable

maternal conditions (infections, hypertensive disorders, asphyxia, etc.), the uptake of evidence-based interventions during ANC could help curb the worldwide burden of both stillbirths and neonatal deaths (Bhutta et al., 2011; Ota et al., 2020).

Considering this reality, it is crucial to assess whether LMICs can effectively adopt and implement the updated guidelines to increase the utilization of ANC interventions. Some frontline actors in sub-Saharan Africa suggest that the model is not feasible within their settings (Mchenga et al., 2019), and others express concerns about cost-effectiveness and maintaining high quality of care across additional visits. Because of this, some authors have proposed a need for a list of the most cost-effective interventions to enable resource-limited settings to prioritize appropriately, even as they seek to mobilize the resources necessary to remodel their health systems toward providing these most ambitious ANC services (Weeks & Temmerman, 2016). Additionally, in LMIC settings, there are future prospects to adopt innovative ANC strategies that have improved the quality and uptake of ANC in HIC settings, such as group ANC models or, where possible, leveraging digital technologies to facilitate antenatal contacts (Chirwa et al., 2020; Palmer et al., 2021; Wu et al., 2020).

Intrapartum Care

The intrapartum period has the potential to provide a triple return on investment by reducing maternal, stillbirth, and neonatal death rates (Bhutta et al., 2011, 2014). Just as for ANC in 2016, and in a bid to achieve some consensus on intrapartum care, the WHO proposed an intrapartum care model in 2018, which aims above all to ensure that the nonnegotiable elements of good delivery care are promoted (Figure 1). This model prioritizes evidence-based practices, but does so while making sure that these practices are locally adaptable worldwide and delivered in ways that are sensitive to the expressed needs of women during labor and childbirth (Kinney et al., 2016; Koblinsky et al., 2016; WHO, 2018c). In line with this, women's expressed desire for a positive childbirth experience entails "giving birth to a healthy baby in a clinically and psychologically safe environment, with continuous emotional support from a birth companion and technically competent clinical staff" (Downe et al., 2018, p. 6; Tunçalp et al., 2015).

The WHO's 58 evidence-based recommendations for intrapartum care are undergirded by the WHO's guiding principles for these recommendations, which include (Oladapo, Tunçalp, et al., 2018; WHO, 2018c):

- Labor and childbirth should be individualized and woman-centered.
- No intervention should be implemented without a clear medical indication.
- Only interventions that serve an immediate purpose and have been proven to be beneficial should be promoted.
- A clear objective that a positive childbirth experience for the woman, the newborn, and her family should be at the forefront of labor and childbirth care at all times.

It is important for healthcare system leaders worldwide to realize that none of the 58 recommendations can stand alone to ensure that women have a positive childbirth experience. Rather, the aim should be to implement the entirety of these guidelines synergistically, in ways that place the woman and her baby at the center of all considerations, that respect the specificity of different health systems, and that do not disrupt the priorities of the most fragile health systems worldwide.

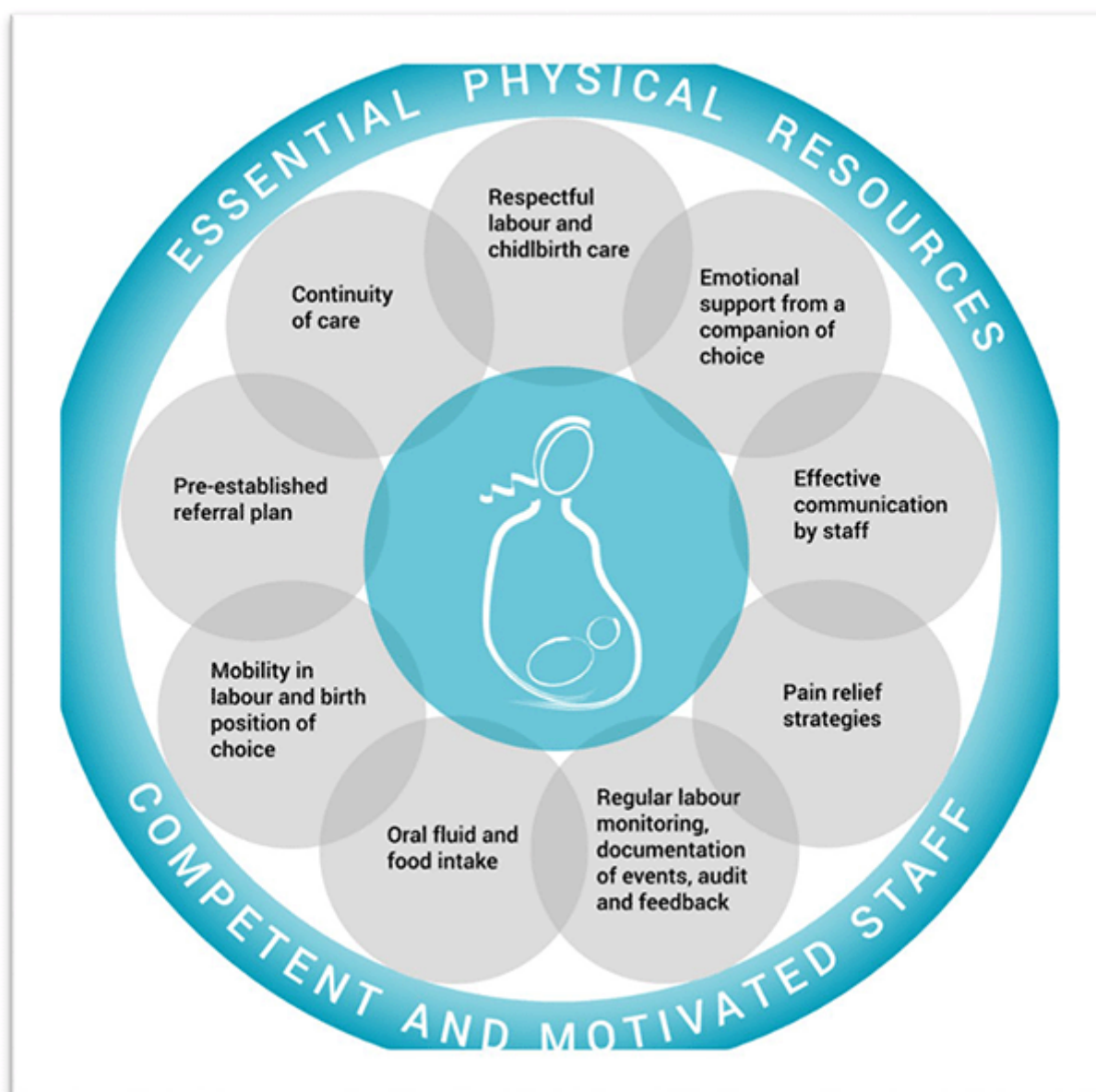


Figure 1. Schematic representation of the World Health Organization intrapartum care model.

Source: Courtesy Oladapo, Tunçalp, et al. (2018).

Postpartum Care

The importance of the postnatal period cannot be overemphasized for several reasons. First, it ushers the mother, her partner, family, and community into parenthood, enabling them to build the bonds essential for the health and well-being of the mother-baby duad (Downe et al., 2018; Finlayson et al., 2020; WHO, 2022c). Next, in terms of care, the postnatal period enables the mother-newborn duad to stay in contact with the health system for continuity of essential care.

Thus, immediate PNC is most likely to be facility-based and entail close monitoring of key clinical indicators to ensure the health and well-being of the mother and the newborn following the stresses of labor and childbirth. Early PNC will seek to foster health and well-being for the mother-baby duad within the confines of their community in the absence of any postpartum complications (NICE, 2021; WHO, 2022c). PNC provides an opportunity for health workers to reinforce counseling regarding optimal nutrition, including breastfeeding, as well as family planning practices. It also enables screening for maternal or newborn complications, including perinatal mood and anxiety disorders (NICE, 2021; WHO, 2022c).

Furthermore, the postnatal period presents specific risks related to morbidity and mortality for mothers and their newborns. Data from LMIC settings indicates that 40% of women experience postpartum complications and that 15% experience severe and life-threatening morbidity (Rahman et al., 2011). In addition, two-thirds of maternal deaths occur in the postnatal period (Alkema et al., 2016; Kassebaum et al., 2014). Approximately three million newborn deaths are reported globally every year, with these deaths accounting for about 45% of deaths of children below 5 years old, and an estimated global rate of 17 deaths per 1,000 live births in 2019 (Sacks & Langlois, 2016; WHO, 2022a). Both maternal and newborn outcomes can be significantly improved by ensuring the availability of and access to high-quality PNC services (Kruk et al., 2018).

Key global strategies, such as the Global Strategy for Women's, Children's, and Adolescents' Health 2016–2030, emphasize the importance of PNC as a mortality- and morbidity-prevention tool. Furthermore, the global health community is increasingly identifying the importance of PNC as an inextricable component of the maternal, newborn, and child care continuum, while validating its importance in ending preventable maternal and newborn deaths (Sacks et al., 2022; WHO, 2018a).

In 2014, the WHO recommended integrated PNC of the mother-newborn duad, indicating a first health check within the first 24 hours after birth, and three additional consultations for the newborn on day 3, between day 7 and 14, and 6 weeks after birth (WHO, 2014). The recommendations further indicated that care must be provided in the facility for at least 24 hours following delivery. In terms of the content of care for the newborn, the recommendations requested an assessment for any abnormal signs, counseling on exclusive breastfeeding, and instructions regarding appropriate umbilical cord care. Regarding maternal care, a complete assessment was indicated as well as recommendations to pursue iron and folic acid supplementation, counseling, and psychosocial support (WHO, 2014).

While these recommendations were backed by evidence, data suggest they did not sufficiently address women's concerns in the postnatal period. Multiple qualitative analyses have shown that women's concerns include self-esteem, competence, and autonomy; adapting to changed intimate and family relationships; and ensuring the health and well-being for both their baby and themselves (Finlayson et al., 2020; Munabi-Babigumira et al., 2021; Sacks et al., 2022). These findings informed the positive-experience revolution, which constituted the backbone of the revised recommendations on PNC for mothers and newborns (WHO, 2022c).

In the revised recommendations, the WHO recommends that all women and newborns receive PNC in the first 24 hours following childbirth, regardless of where the birth occurs, and subsequent postnatal checkups in the first six weeks (WHO, 2022c). The guideline proposed 63 evidence-based recommendations on PNC, with 31 new recommendations that were not contained within the previous guideline. To ensure a positive postnatal experience, these guidelines promote maternal and newborn health assessments aimed at the early detection of conditions that may adversely affect their health and well-being postpartum, women's capacity to care for themselves and their newborns, and the development of the newborn.

A key difference between these recommendations and the previous recommendations (as well as other postnatal care recommendations) is the proposal of health system interventions. These health interventions identified by the WHO's guideline development group address broader health system structures, processes, and policies, as well as care delivery innovations, to improve the uptake of services and women's experience following care (NICE, 2021; WHO, 2014, 2022c).

Conclusions

Maternal health is a global health policy priority. While criticism remains that focusing on pregnancy and birth-related risks does not sufficiently address the broader concept of women's health, the fight against preventable maternal mortality and morbidity remains a key guiding principle in the efforts aimed at improving access to care for women and adolescent girls worldwide, especially in resource-limited settings. Yearly, more than 210 million women and girls become pregnant, for an expected 140 million births, and the improvement of maternal health is imperative.

Both declining maternal mortality rates in HICs and stagnant rates in LMICs necessitate global policies that look beyond survival and center instead on maternal well-being and high-quality care for women worldwide. Quantitative measures other than maternal mortality rates, such as severe maternal morbidity, in LMICs are likely to show maternal health inequities to be greater than perceived. In addition, monitoring for the burden of disease beyond pregnancy, such as increased lifelong risk factors of cardiovascular disease, including hypertension, and diabetes after exposure in pregnancy further reinforces the impact maternal health has on the well-being of women outside of pregnancy. Furthermore, with perinatal mood and anxiety disorders (PMADs) constituting the highest percentage of mental disorders in women, addressing PMADs remains a key aspect of improving the mental health of women.

Existing indicators aimed at evaluating maternal health and well-being—especially in high-burden countries—focus essentially on the quantitative aspects of care. Yet some of the most important aspects of care are challenging to assess quantitatively. It is difficult to quantify the quality of care women receive or to measure how much consideration is given to respecting their rights while delivering maternal health services, both globally and in LMICs. Therefore, it is important to adopt a rights-based equity lens when analyzing the quantity, quality, and efficiency of maternal healthcare services worldwide, even in HICs, which manifest better maternal and newborn health outcomes compared to their LMIC counterparts.

Improving and investing in high-quality maternal health services provides significant intergeneration impacts that are likely to benefit both women and children. Every aspect of the maternal experience, from preconception to postnatal, provides opportunities for interventions aimed at improving the overall well-being of women and children. Initiatives aimed at reducing barriers in maternal health must look beyond medical care and incorporate processes that minimize personal, social, political, financial, and cultural barriers to maternal health. High-quality maternal healthcare must incorporate all aspects that provide women the opportunity to enter and leave the birth experience with safety, dignity, and autonomy.

Further Reading

Ceschia, A., & Horton, R. (2016). Maternal health: Time for a radical reappraisal. *The Lancet*, 388(10056), 2064–2066.

Cleland, J., Bernstein, S., Ezeh, A., Faundes, A., Glasier, A., & Innis, J. (2016). Family planning: The unfinished agenda <[https://doi.org/10.1016/S0140-6736\(06\)69480-4](https://doi.org/10.1016/S0140-6736(06)69480-4)>. *The Lancet*, 388(10041), 1810–1827. [https://doi.org/10.1016/S0140-6736\(16\)00278-2](https://doi.org/10.1016/S0140-6736(16)00278-2) <[https://doi.org/10.1016/S0140-6736\(16\)00278-2](https://doi.org/10.1016/S0140-6736(16)00278-2)>

Firoz, T., Pineles, B., Navrange, N., Grimshaw, A., Oladapo, O., & Chou, D. (2022). Non communicable diseases and maternal health: A scoping review. *BMC Pregnancy and Childbirth*, 22(1), 1–14.

IPPF. (2019). *Welcome to ICPD25 Commitments!* <<https://www.icpd25commitments.org/>>

Kassebaum, N. J., & GBD 2015 Maternal Mortality Collaborators. (2016). Global, regional, and national levels of maternal mortality, 1990–2015: A systematic analysis for the Global Burden of Disease Study 2015 <[https://doi.org/10.1016/S0140-6736\(16\)31470-2](https://doi.org/10.1016/S0140-6736(16)31470-2)>. *The Lancet*, 388(10053), 1775–1812.

Kasthurirathne, S. N., Mamlin, B. W., Purkayastha, S., & Cullen, T. (2017). Overcoming the maternal care crisis: How can lessons learnt in global health informatics address US maternal health outcomes? *AMIA Annual Symposium Proceedings*, 2017, 1034–1043.

Knaul, F. M., Langer, A., Atun, R., Rodin, D., Frenk, J., & Bonita, R. (2016). Rethinking maternal health <[https://doi.org/10.1016/s2214-109x\(16\)00044-9](https://doi.org/10.1016/s2214-109x(16)00044-9)>. *The Lancet Global Health*, 4(4), e227–e228.

Ronsmans, C., Borghi, J., Campbell, O., Filippi, V., Graham, W., Koblinsky, M., & Mills, A. (2015). Maternal mortality: Who, when, where, and why <[https://doi.org/10.1016/S0140-6736\(06\)69380-X](https://doi.org/10.1016/S0140-6736(06)69380-X)>. *The Lancet*, 368(9542), 1189–1200.

Say, L., Chou, D., Gemmil, A., Tunçalp, O., Moller, A., Daniels, J., Gülmezoglu, A. M., Temmerman, M., & Alkema, L. (2014). Global causes of maternal death: A WHO systematic analysis [https://doi.org/10.1016/S2214-109X\(14\)70227-X](https://doi.org/10.1016/S2214-109X(14)70227-X). *The Lancet Global Health*, 2(6), e323–e333.

Singh, S., Remez, L., Sedgh, G., Kwok, L., & Onda, T. (2018). *Abortion worldwide 2017: Uneven progress and unequal access*. Guttmacher Institute.

Temmerman, M., Khosla, R., Bhutta, Z. A., & Bustreo, F. (2020). Toward a new global strategy for women's, children's and adolescents' health <https://pubmed.ncbi.nlm.nih.gov/26371228/>. *BMJ Global Health*, 351, h4414.

References

Abekah-Nkrumah, G. (2019). Trends in utilisation and inequality in the use of reproductive health services in sub-Saharan Africa <https://doi.org/10.1186/s12889-019-7865-z>. *BMC Public Health*, 19, 1541.

Actis Danna, V., Bedwell, C., Wakasiaka, S., & Lavender, T. (2020). Utility of the three-delays model and its potential for supporting a solution-based approach to accessing intrapartum care in low- and middle-income countries: A qualitative evidence synthesis <https://doi.org/10.1080/16549716.2020.1819052>. *Global Health Action*, 13(1), 1819052.

Alderdice, F., & Newham, J. (2016). Global maternal mental health: Where you live matters <https://doi.org/10.1080/02646838.2016.1126945>. *Journal of Reproductive and Infant Psychology*, 34(1), 1–2.

Alkema, L., Chou, D., Hogan, D., Zhang, S., Moller, A.-B., Gemmill, A., Fat, D. M., Boerma, T., Temmerman, M., Mathers, C., & Say, L. (2016). Global, regional, and national levels and trends in maternal mortality between 1990 and 2015, with scenario-based projections to 2030: A systematic analysis by the UN Maternal Mortality Estimation Inter-Agency Group <https://doi.org/10.1097/01.goa.0000504718.91214.7d>. *Obstetric Anesthesia Digest*, 36(4), 191.

American College of Obstetrics and Gynecology, Kilpatrick, S. K., & Ecker, J. L. (2016). Severe maternal morbidity: Screening and review <https://doi.org/10.1016/j.ajog.2016.07.050>. *American Journal of Obstetrics and Gynecology*, 215(3), B17–B22.

Amouzou, A., Hazel, E., Vaz, L., Sanni, Y., & Moran, A. (2020). Discordance in postnatal care between mothers and newborns: Measurement artifact or missed opportunity? <https://doi.org/10.7189/jogh.10.010505> *Journal of Global Health*, 10(1), 010505.

Asaolu, I. O., Alaofè, H., Gunn, J. K. L., Adu, A. K., Monroy, A. J., Ehiri, J. E., Hayden, M. H., & Ernst, K. C. (2018). Measuring women's empowerment in sub-Saharan Africa: Exploratory and confirmatory factor analyses of the demographic and health surveys <https://doi.org/10.3389/fpsyg.2018.00994>. *Frontiers in Psychology*, 9, 994.

Atif, N., Lovell, K., & Rahman, A. (2015). Maternal mental health: The missing “m” in the global maternal and child health agenda <https://doi.org/10.1053/j.semperi.2015.06.007>. *Seminars in Perinatology*, 39(5), 345–352.

Bain, L. E., Aboagye, R. G., Dowou, R. K., Kongnyuy, E. J., Memiah, P., & Amu, H. (2022). Prevalence and determinants of maternal healthcare utilisation among young women in sub-Saharan Africa: Cross-sectional analyses of demographic and health survey data <https://doi.org/10.1186/s12889-022-13037-8>. *BMC Public Health*, 22(1), 647.

- Baldo, M. H. (2001). The antenatal care debate. *Eastern Mediterranean Health Journal* [La Revue De Sante De La Mediterranee Orientale], 7(6), 1046–1055.
- Barker, M., Dombrowski, S. U., Colbourn, T., Fall, C. H. D., Kriznik, N. M., Lawrence, W. T., Norris, S. A., Ngaiza, G., Patel, D., Skordis-Worrall, J., Sniehotta, F. F., Steegers-Theunissen, R., Vogel, C., Woods-Townsend, K., & Stephenson, J. (2018). Intervention strategies to improve nutrition and health behaviours before conception [https://doi.org/10.1016/S0140-6736\(18\)30313-1](https://doi.org/10.1016/S0140-6736(18)30313-1). *The Lancet*, 391(10132), 1853–1864.
- Beattie, R. M., Brown, N. J., & Cass, H. (2015). Millennium Development Goals progress report <https://doi.org/10.1136/archdischild-2014-307933>. *Archives of Disease in Childhood*, 100, s1.
- Bennett, I., Schott, W., Krutikova, S., & Behrman, J. R. (2016). Maternal mental health and child growth and development in four low and middle income countries <https://doi.org/10.1136/jech-2014-205311>. *Journal of Epidemiology and Community Health*, 70(2), 168–173.
- Benova, L., Owolabi, O., Radovich, E., Wong, K. L. M., Macleod, D., Langlois, E. V., & Campbell, O. M. R. (2019). Provision of postpartum care to women giving birth in health facilities in sub-Saharan Africa: A cross-sectional study using Demographic and Health Survey data from 33 countries <https://doi.org/10.1371/journal.pmed.1002943>. *PLOS Medicine*, 16(10), e1002943.
- Bergström, S. (2016). Global maternal health and newborn health: Looking backwards to learn from history <https://doi.org/10.1016/j.bpobgyn.2016.05.010>. *Best Practice & Research Clinical Obstetrics & Gynaecology*, 36, 3–13.
- Betrán, A. P., Ye, J., Moller, A.-B., Zhang, J., Gülmezoglu, A. M., & Torloni, M. R. (2016). The increasing trend in caesarean section rates: Global, regional and national estimates: 1990–2014 <https://doi.org/10.1371/journal.pone.0148343>. *PLOS ONE*, 11(2), e0148343.
- Bhutta, Z. A., Das, J. K., Bahl, R., Lawn, J. E., Salam, R. A., Paul, V. K., Sankar, M. J., Blencowe, H., Rizvi, A., Chou, V. B., & Walker, N. (2014). Can available interventions end preventable deaths in mothers, newborn babies, and stillbirths, and at what cost? [https://doi.org/10.1016/S0140-6736\(14\)60792-3](https://doi.org/10.1016/S0140-6736(14)60792-3) *The Lancet*, 384(9940), 348–370.
- Bhutta, Z. A., Karmaliani, R., & Ali, S. (2019). Men engagement in women empowerment for improving maternal mental health through cash transfer and life skills building (MEWE): *Clinical trial on empowerment: Life skills building training—Clinical Trials Registry—ICH GCP* <https://ichgcp.net/clinical-trials-registry/NCT04125251>.
- Bhutta, Z. A., Yakoob, M. Y., Lawn, J. E., Rizvi, A., Friberg, I. K., Weissman, E., Buchmann, E., Goldenberg, R. L., & Lancet's Stillbirths Series Steering Committee. (2011). Stillbirths: What difference can we make and at what cost? [https://doi.org/10.1016/S0140-6736\(10\)62269-6](https://doi.org/10.1016/S0140-6736(10)62269-6) *The Lancet*, 377(9776), 1523–1538.
- Bick, D., Silverio, S. A., Bye, A., & Chang, Y.-S. (2020). Postnatal care following hypertensive disorders of pregnancy: A qualitative study of views and experiences of primary and secondary care clinicians <https://doi.org/10.1136/bmjopen-2019-034382>. *BMJ Open*, 10(1), e034382.
- Boerma, T., Ronsmans, C., Melesse, D. Y., Barros, A. J. D., Barros, F. C., Juan, L., Moller, A.-B., Say, L., Hosseinpoor, A. R., Yi, M., de Lyra Rabello Neto, D., & Temmerman, M. (2018). Global epidemiology of use of and disparities in caesarean sections [https://doi.org/10.1016/S0140-6736\(18\)31928-7](https://doi.org/10.1016/S0140-6736(18)31928-7). *The Lancet*, 392(10155), 1341–1348.

- Bongaarts, J. (2016). WHO, UNICEF, UNFPA, World Bank Group, and United Nations population division trends in maternal mortality: 1990 to 2015 Geneva; World Health Organization, 2015 [_<https://doi.org/10.1111/padr.12033>_.](https://doi.org/10.1111/padr.12033) *Population and Development Review*, 42(4), 726.
- Brizuela, V., & Tunçalp, Ö. (2017). Global initiatives in maternal and newborn health [_<https://doi.org/10.1177/1753495X16684987>_.](https://doi.org/10.1177/1753495X16684987) *Obstetric Medicine*, 10(1), 21–25.
- Brown, W., Druce, N., Bunting, J., Radloff, S., Koroma, D., Gupta, S., Siems, B., Kerrigan, M., Kress, D., & Darmstadt, G. L. (2014). Developing the “120 by 20” goal for the Global FP2020 Initiative [_<https://doi.org/10.1111/j.1728-4465.2014.00377.x>_.](https://doi.org/10.1111/j.1728-4465.2014.00377.x) *Studies in Family Planning*, 45(1), 73–84.
- Buchmann, E. J., Stones, W., & Thomas, N. (2016). Preventing deaths from complications of labour and delivery [_<https://doi.org/10.1016/j.bpobgyn.2016.05.012>_.](https://doi.org/10.1016/j.bpobgyn.2016.05.012) *Best Practice & Research Clinical Obstetrics & Gynaecology*, 36, 103–115.
- Cahill, N., Sonneveldt, E., Stover, J., Weinberger, M., Williamson, J., Wei, C., Brown, W., & Alkema, L. (2018). Modern contraceptive use, unmet need, and demand satisfied among women of reproductive age who are married or in a union in the focus countries of the Family Planning 2020 initiative: A systematic analysis using the Family Planning Estimation Tool [_<https://doi.org/10.1016/S0140-6736\(17\)33104-5>_.](https://doi.org/10.1016/S0140-6736(17)33104-5) *The Lancet*, 391(10123), 870–882.
- Population and Development Review. Cairo 25: The UN Population Commission on Population Change since the 1994 ICPD [_<https://doi.org/10.1111/padr.12258>_.](https://doi.org/10.1111/padr.12258) (2019). *Population and Development Review*, 45(2), 445–448.
- Camacho, E. M., & Shields, G. E. (2018). Cost-effectiveness of interventions for perinatal anxiety and/or depression: A systematic review [_<https://doi.org/10.1136/bmjopen-2018-022022>_.](https://doi.org/10.1136/bmjopen-2018-022022) *BMJ Open*, 8(8), e022022.
- Campbell, O. M. R., Graham, W. J., & Lancet Maternal Survival Series Steering Group. (2006). Strategies for reducing maternal mortality: Getting on with what works [_<https://doi.org/10.1016/S0140-6736\(06\)69381-1>_.](https://doi.org/10.1016/S0140-6736(06)69381-1) *The Lancet*, 368(9543), 1284–1299.
- Carniato, G. (2021). Access to maternal health care for indigenous Australians under international law. *Journal of Law and Medicine*, 28(4), 1018–1034.
- Carroli, G., Rooney, C., & Villar, J. (2001). How effective is antenatal care in preventing maternal mortality and serious morbidity? An overview of the evidence [_<https://doi.org/10.1046/j.1365-3016.2001.0150s1001.x>_.](https://doi.org/10.1046/j.1365-3016.2001.0150s1001.x) *Paediatric and Perinatal Epidemiology*, 15(S1), 1–42.
- Carroli, G., Villar, J., Piaggio, G., Khan-Neelofur, D., Gülmezoglu, M., Mugford, M., Lumbiganon, P., Farnot, U., & Bersgjø, P. (2001). WHO systematic review of randomised controlled trials of routine antenatal care [_<https://doi.org/10.1016/S0140-6736\(00\)04723-1>_.](https://doi.org/10.1016/S0140-6736(00)04723-1) *The Lancet*, 357(9268), 1565–1570.
- Ceschia, A., & Horton, R. (2016). Maternal health: Time for a radical reappraisal [_<https://doi.org/10.1016/S0140-6736\(16\)31534-3>_.](https://doi.org/10.1016/S0140-6736(16)31534-3) *The Lancet*, 388(10056), 2064–2066.
- Cheng, Y., Xu, X., Xu, J., Wuillaume, F., Zhu, J., Gibson, D., & Temmerman, M. (2008). The need for integrating family planning and postabortion care in China [_<https://doi.org/10.1016/j.ijgo.2008.06.013>_.](https://doi.org/10.1016/j.ijgo.2008.06.013) *International Journal of Gynecology & Obstetrics*, 103(2), 140–143.

- Chin, K., Wendt, A., Bennett, I. M., & Bhat, A. (2022). Suicide and maternal mortality [_<https://doi.org/10.1007/s11920-022-01334-3>](https://doi.org/10.1007/s11920-022-01334-3). *Current Psychiatry Reports*, 24(4), 239–275.
- Chirwa, E., Kapito, E., Jere, D. L., Kafulafula, U., Chodzaza, E., Chorwe-Sungani, G., Gresh, A., Liu, L., Abrams, E. T., Klima, C. S., McCreary, L. L., Norr, K. F., & Patil, C. L. (2020). An effectiveness-implementation hybrid type 1 trial assessing the impact of group versus individual antenatal care on maternal and infant outcomes in Malawi [_<https://doi.org/10.1186/s12889-020-8276-x>](https://doi.org/10.1186/s12889-020-8276-x). *BMC Public Health*, 20(1), 205.
- Chmielewska, B., Barratt, I., Townsend, R., Kalafat, E., van der Meulen, J., Gurol-Urganci, I., O'Brien, P., Morris, E., Draycott, T., Thangaratinam, S., Le Doare, K., Ladhani, S., von Dadelszen, P., Magee, L., & Khalil, A. (2021). Effects of the COVID-19 pandemic on maternal and perinatal outcomes: A systematic review and meta-analysis [_<https://doi.org/10.1016/S2214-109X\(21\)00079-6>](https://doi.org/10.1016/S2214-109X(21)00079-6). *The Lancet Global Health*, 9(6), e759–e772.
- Cusimano, M. C., Pudwell, J., Roddy, M., Cho, C.-K. J., & Smith, G. N. (2014). The maternal health clinic: An initiative for cardiovascular risk identification in women with pregnancy-related complications [_<https://doi.org/10.1016/j.ajog.2013.12.001>](https://doi.org/10.1016/j.ajog.2013.12.001). *American Journal of Obstetrics and Gynecology*, 210(5), 438.e1–438.e9.
- da Silva, A. A. M. (2019). Monitoring trends in socioeconomic, maternal and child health inequalities [_<https://doi.org/10.1093/ije/dyz024>](https://doi.org/10.1093/ije/dyz024). *International Journal of Epidemiology*, 48(Suppl. 1), i1–i3.
- de Masi, S., Bucagu, M., Tunçalp, Ö., Peña-Rosas, J. P., Lawrie, T., Oladapo, O. T., & Gülmezoglu, M. (2017). Integrated person-centered health care for all women during pregnancy: Implementing World Health Organization recommendations on antenatal care for a positive pregnancy experience [_<https://doi.org/10.9745/GHSP-D-17-00141>](https://doi.org/10.9745/GHSP-D-17-00141). *Global Health: Science and Practice*, 5(2), 197–201.
- De-Regil, L. M., Fernández-Gaxiola, A. C., Dowswell, T., & Peña-Rosas, J. P. (2010). Effects and safety of periconceptional folate supplementation for preventing birth defects [_<https://doi.org/10.1002/14651858.CD007950.pub2>](https://doi.org/10.1002/14651858.CD007950.pub2). *The Cochrane Database of Systematic Reviews*, 10, i-102.
- Dhaher, E., Mikolajczyk, R. T., Maxwell, A. E., & Krämer, A. (2008). Factors associated with lack of postnatal care among Palestinian women: A cross-sectional study of three clinics in the West Bank [_<https://doi.org/10.1186/1471-2393-8-26>](https://doi.org/10.1186/1471-2393-8-26). *BMC Pregnancy and Childbirth*, 8, 26.
- Downe, S., Finlayson, K., Oladapo, O., Bonet, M., & Gülmezoglu, A. M. (2018). What matters to women during childbirth: A systematic qualitative review [_<https://doi.org/10.1371/journal.pone.0194906>](https://doi.org/10.1371/journal.pone.0194906). *PLOS ONE*, 13(4), e0194906.
- Downe, S., Finlayson, K., Tunçalp, Ö., & Metin Gülmezoglu, A. (2016). What matters to women: A systematic scoping review to identify the processes and outcomes of antenatal care provision that are important to healthy pregnant women [_<https://doi.org/10.1111/1471-0528.13819>](https://doi.org/10.1111/1471-0528.13819). *BJOG: An International Journal of Obstetrics & Gynaecology*, 123(4), 529–539.
- Endler, M., Al-Haidari, T., Benedetto, C., Chowdhury, S., Christilaw, J., El Kak, F., Galimberti, D., Gutierrez, M., Ibrahim, S., Kumari, S., McNicholas, C., Flores, D. M., Muganda, J., Ramirez-Negrin, A., Senanayake, H., Sohail, R., Temmerman, M., & Gemzell Danielsson, K. (2022). Are sexual and reproductive health and rights taught in medical school? Results from a global survey [_<https://doi.org/10.1002/ijgo.14339>](https://doi.org/10.1002/ijgo.14339). *International Journal of Gynecology & Obstetrics*, 159(3), 735–752.

- Ewerling, F., Lynch, J. W., Victora, C. G., van Eerdewijk, A., Tysler, M., & Barros, A. J. D. (2017). The SWPER index for women's empowerment in Africa: Development and validation of an index based on survey data [_<https://doi.org/10.1016/S2214-109X\(17\)30292-9>_](https://doi.org/10.1016/S2214-109X(17)30292-9). *The Lancet Global Health*, 5(9), e916–e923.
- Family Planning 2020. (2022). *The Arc of Progress* [_<http://progress.familyplanning2020.org/arc-of-progress>_](http://progress.familyplanning2020.org/arc-of-progress).
- Finlayson, K., Crossland, N., Bonet, M., & Downe, S. (2020). What matters to women in the postnatal period: A meta-synthesis of qualitative studies [_<https://doi.org/10.1371/journal.pone.0231415>_](https://doi.org/10.1371/journal.pone.0231415). *PLOS ONE*, 15(4), e0231415.
- Fleming, T. P., Watkins, A. J., Velazquez, M. A., Mathers, J. C., Prentice, A. M., Stephenson, J., Barker, M., Saffery, R., Yajnik, C. S., Eckert, J. J., Hanson, M. A., Forrester, T., Gluckman, P. D., & Godfrey, K. M. (2018). Origins of lifetime health around the time of conception: Causes and consequences [_<https://doi.org/10.1016/S0140-6736\(18\)30312-X>_](https://doi.org/10.1016/S0140-6736(18)30312-X). *The Lancet*, 391(10132), 1842–1852.
- Gibbons, L., Belizan, J. M., Lauer, J. A., Betran, A. P., Merialdi, M., & Althabe, F. (2012). Inequities in the use of cesarean section deliveries in the world [_<https://doi.org/10.1016/j.ajog.2012.02.026>_](https://doi.org/10.1016/j.ajog.2012.02.026). *American Journal of Obstetrics and Gynecology*, 206(4), 331.e1–331.e19.
- Graham, W., Woodd, S., Byass, P., Filippi, V., Gon, G., Virgo, S., Chou, D., Hounton, S., Lozano, R., Pattinson, R., & Singh, S. (2016). Diversity and divergence: The dynamic burden of poor maternal health [_<https://doi.org/10.1016/S0140-6736\(16\)31533-1>_](https://doi.org/10.1016/S0140-6736(16)31533-1). *The Lancet*, 388(10056), 2164–2175.
- GrantWatch. (2021). Maternal health: What funders have been supporting [_<https://doi.org/10.1377/hlthaff.2021.00350>_](https://doi.org/10.1377/hlthaff.2021.00350). *Health Affairs*, 40(4), 675–676.
- Gupta, M. L., Aborigo, R. A., Adongo, P. B., Rominski, S., Hodgson, A., Engmann, C. M., & Moyer, C. A. (2015). Grandmothers as gatekeepers? The role of grandmothers in influencing health-seeking for mothers and newborns in rural northern Ghana [_<https://doi.org/10.1080/17441692.2014.1002413>_](https://doi.org/10.1080/17441692.2014.1002413). *Global Public Health*, 10(9), 1078–1091.
- Hall, M., Chng, P. K., & Macgillivray, I. (1980). Is routine antenatal care worth while? [_<https://doi.org/10.1016/S0140-6736\(80\)92950-5>_](https://doi.org/10.1016/S0140-6736(80)92950-5) *The Lancet*, 316(8185), 78–80.
- Hardee, K., & Jordan, S. (2021). Advancing rights-based family planning from 2020 to 2030 [_<https://doi.org/10.2147/OAJC.S324678>_](https://doi.org/10.2147/OAJC.S324678). *Open Access Journal of Contraception*, 12, 157–171.
- Hare, M. J. L., Barzi, F., Boyle, J. A., Guthridge, S., Dyck, R. F., Barr, E. L. M., Singh, G., Falhammar, H., Webster, V., Shaw, J. E., & Maple-Brown, L. J. (2020). Diabetes during pregnancy and birthweight trends among aboriginal and non-aboriginal people in the Northern Territory of Australia over 30 years [_<https://doi.org/10.1016/j.lanwpc.2020.100005>_](https://doi.org/10.1016/j.lanwpc.2020.100005). *The Lancet Regional Health—Western Pacific*, 1, 100005.
- Hjort, L., Novakovic, B., Grunnet, L. G., Maple-Brown, L., Damm, P., Desoye, G., & Saffery, R. (2019). Diabetes in pregnancy and epigenetic mechanisms-how the first 9 months from conception might affect the child's epigenome and later risk of disease [_<https://doi.org/10.1016/S2213-8587\(19\)30078-6>_](https://doi.org/10.1016/S2213-8587(19)30078-6). *The Lancet Diabetes & Endocrinology*, 7(10), 796–806.
- Hofmeyr, G. J., & Hodnett, E. D. (2013). Antenatal care packages with reduced visits and perinatal mortality: A secondary analysis of the WHO antenatal care trial—Comentary: Routine antenatal visits for healthy pregnant women do make a difference [_<https://doi.org/10.1186/1742-4755-10-20>_](https://doi.org/10.1186/1742-4755-10-20). *Reproductive Health*, 10(1), 20.

Howard, L. M., Molyneaux, E., Dennis, C.-L., Rochat, T., Stein, A., & Milgrom, J. (2014). Non-psychotic mental disorders in the perinatal period [_<https://doi.org/10.1016/S0140-6736\(14\)61276-9>_.](https://doi.org/10.1016/S0140-6736(14)61276-9) *The Lancet*, 384(9956), 1775–1788.

Howard, L. M., Piot, P., & Stein, A. (2014). No health without perinatal mental health [_<https://doi.org/10.1016/S0140-6736\(14\)62040-7>_.](https://doi.org/10.1016/S0140-6736(14)62040-7) *The Lancet*, 384(9956), 1723–1724.

Huang, Y., Shallcross, D., Pi, L., Tian, F., Pan, J., & Ronsmans, C. (2018). Ethnicity and maternal and child health outcomes and service coverage in western China: A systematic review and meta-analysis [_<https://doi.org/10.1016/S2214-109X\(17\)30445-X>_.](https://doi.org/10.1016/S2214-109X(17)30445-X) *The Lancet Global Health*, 6(1), e39–e56.

International Planned Parenthood Federation. (2019). *Welcome to ICPD25 Commitments!* [_<https://www.icpd25commitments.org/>_](https://www.icpd25commitments.org/)

Iyengar, S. S., & DeVoe, S. E. (2003). Rethinking the value of choice: Considering cultural mediators of intrinsic motivation. *Nebraska Symposium on Motivation*, 49, 129–174.

Jones, I., Chandra, P. S., Dazzan, P., & Howard, L. M. (2014). Bipolar disorder, affective psychosis, and schizophrenia in pregnancy and the post-partum period [_<https://doi.org/10.1016/S0140-6736\(14\)61278-2>_.](https://doi.org/10.1016/S0140-6736(14)61278-2) *The Lancet*, 384(9956), 1789–1799.

Kabeer, N. (1999). Resources, agency, achievements: Reflections on the measurement of women's empowerment [_<https://doi.org/10.1111/1467-7660.00125>_.](https://doi.org/10.1111/1467-7660.00125) *Development and Change*, 30(3), 435–464.

Kasper, A., Mohwinkel, L.-M., Nowak, A. C., & Kolip, P. (2022). Maternal health care for refugee women—A qualitative review [_<https://doi.org/10.1016/j.midw.2021.103157>_.](https://doi.org/10.1016/j.midw.2021.103157) *Midwifery*, 104, 103157.

Kassebaum, N. J., Bertozzi-Villa, A., Coggeshall, M. S., Shackelford, K. A., Steiner, C., Heuton, K. R., Gonzalez-Medina, D., Barber, R., Huynh, C., Dicker, D., Templin, T., Wolock, T. M., Ozgoren, A. A., Abd-Allah, F., Abera, S. F., Abubakar, I., Achoki, T., Adelekan, A., Ademi, Z., . . . Lozano, R. (2014). Global, regional, and national levels and causes of maternal mortality during 1990–2013: A systematic analysis for the Global Burden of Disease Study 2013 [_<https://doi.org/10.1016/S0140-6736\(14\)60696-6>_.](https://doi.org/10.1016/S0140-6736(14)60696-6) *The Lancet*, 384(9947), 980–1004.

Kazembe, A., Simwaka, A., Dougherty, K., Petross, C., Chakhame, B., Chodzaza, E., Chisuse, I., Kamanga, M., Sun, C., & George, M. (2022). Experiences of women with physical disabilities accessing prenatal care in low- and middle-income countries [_<https://doi.org/10.1111/phn.13087>_.](https://doi.org/10.1111/phn.13087) *Public Health Nursing*, 39(05), 1156–1166.

Khosla, R., Say, L., & Temmerman, M. (2015). Sexual health, human rights, and law [_<https://doi.org/10.1016/S0140-6736\(15\)61449-0>_.](https://doi.org/10.1016/S0140-6736(15)61449-0) *The Lancet*, 386(9995), 725–726.

Kinney, M. V., Boldosser-Boesch, A., & McCallon, B. (2016). Quality, equity, and dignity for women and babies [_<https://doi.org/10.1016/S0140-6736\(16\)31525-2>_.](https://doi.org/10.1016/S0140-6736(16)31525-2) *The Lancet*, 388(10056), 2066–2068.

Knaul, F. M., Langer, A., Atun, R., Rodin, D., Frenk, J., & Bonita, R. (2016). Rethinking maternal health [_<https://doi.org/10.1016/S2214-109X\(16\)00044-9>_.](https://doi.org/10.1016/S2214-109X(16)00044-9) *The Lancet Global Health*, 4(4), e227–e228.

Koblinsky, M., Moyer, C. A., Calvert, C., Campbell, J., Campbell, O. M. R., Feigl, A. B., Graham, W. J., Hatt, L., Hodgins, S., Matthews, Z., McDougall, L., Moran, A. C., Nandakumar, A. K., & Langer, A. (2016). Quality maternity care for every woman, everywhere: A call to action [_<https://doi.org/10.1016/S0140-6736\(16\)31333-2>_.](https://doi.org/10.1016/S0140-6736(16)31333-2) *The Lancet*, 388(10057), 2307–2320.

- Kotlar, B., Gerson, E., Petrillo, S., Langer, A., & Tiemeier, H. (2021). The impact of the COVID-19 pandemic on maternal and perinatal health: A scoping review [<https://doi.org/10.1186/s12978-021-01070-6>](https://doi.org/10.1186/s12978-021-01070-6). *Reproductive Health*, 18(1), 10.
- Kruk, M. E., Gage, A. D., Arsenault, C., Jordan, K., Leslie, H. H., Roder-DeWan, S., Adeyi, O., Barker, P., Daelmans, B., Doubova, S. V., English, M., García-Elorrio, E., Guanais, F., Gureje, O., Hirschhorn, L. R., Jiang, L., Kelley, E., Lemango, E. T., Liljestrand, J., . . . Pate, M. (2018). High-quality health systems in the Sustainable Development Goals era: Time for a revolution [<https://doi.org/10.1016/S2214-109X\(18\)30386-3>](https://doi.org/10.1016/S2214-109X(18)30386-3). *The Lancet Global Health*, 6(11), e1196–e1252.
- Kujawski, S., Mbaruku, G., Freedman, L. P., Ramsey, K., Moyo, W., & Kruk, M. E. (2015). Association between disrespect and abuse during childbirth and women's confidence in health facilities in Tanzania [<https://doi.org/10.1007/s10995-015-1743-9>](https://doi.org/10.1007/s10995-015-1743-9). *Maternal and Child Health Journal*, 19(10), 2243–2250.
- Langer, A., Meleis, A., Knaul, F. M., Atun, R., Aran, M., Arreola-Ornelas, H., Bhutta, Z. A., Binagwaho, A., Bonita, R., Caglia, J. M., Claeson, M., Davies, J., Donnay, F. A., Gausman, J. M., Glickman, C., Kearns, A. D., Kendall, T., Lozano, R., Seboni, N., . . . Frenk, J. (2015). Women and health: The key for sustainable development [<https://doi.org/10.1016/S0140-6736\(15\)60497-4>](https://doi.org/10.1016/S0140-6736(15)60497-4). *The Lancet*, 386(9999), 1165–1210.
- Langlois, É. V., Miskurka, M., Zunzunegui, M. V., Ghaffar, A., Ziegler, D., & Karp, I. (2015). Inequities in postnatal care in low- and middle-income countries: A systematic review and meta-analysis [<https://doi.org/10.2471/BLT.14.140996>](https://doi.org/10.2471/BLT.14.140996). *Bulletin of the World Health Organization*, 93(4), 259–270G.
- Lassi, Z. S., Dean, S. V., Mallick, D., & Bhutta, Z. A. (2014). Preconception care: Delivery strategies and packages for care [<https://doi.org/10.1186/1742-4755-11-S3-S7>](https://doi.org/10.1186/1742-4755-11-S3-S7). *Reproductive Health*, 11(3), S7.
- Lassi, Z. S., Imam, A. M., Dean, S. V., & Bhutta, Z. A. (2014). Preconception care: Preventing and treating infections [<https://doi.org/10.1186/1742-4755-11-S3-S4>](https://doi.org/10.1186/1742-4755-11-S3-S4). *Reproductive Health*, 11(3), S4.
- Lassi, Z. S., Majeed, A., Rashid, S., Yakoob, M. Y., & Bhutta, Z. A. (2013). The interconnections between maternal and newborn health—Evidence and implications for policy [<https://doi.org/10.3109/14767058.2013.784737>](https://doi.org/10.3109/14767058.2013.784737). *The Journal of Maternal-Fetal & Neonatal Medicine*, 26(Suppl. 1), 3–53.
- Lawn, J. E., Blencowe, H., Pattinson, R., Cousens, S., Kumar, R., Ibiebele, I., Gardosi, J., Day, L. T., Stanton, C., & Lancet's Stillbirths Series Steering Committee. (2011). Stillbirths: Where? When? Why? How to make the data count? [<https://doi.org/10.1016/S0140-6736\(10\)62187-3>](https://doi.org/10.1016/S0140-6736(10)62187-3) *The Lancet*, 377(9775), 1448–1463.
- Lawn, J. E., Blencowe, H., Waiswa, P., Amouzou, A., Mathers, C., Hogan, D., Flenady, V., Frøen, J. F., Qureshi, Z. U., Calderwood, C., Shiekh, S., Jassir, F. B., You, D., McClure, E. M., Mathai, M., Cousens, S., Lancet Ending Preventable Stillbirths Series Study Group, & Lancet Stillbirth Epidemiology Investigator Group. (2016). Stillbirths: Rates, risk factors, and acceleration towards 2030 [<https://doi.org/10.1016/S0140-6736\(15\)00837-5>](https://doi.org/10.1016/S0140-6736(15)00837-5). *The Lancet*, 387(10018), 587–603.
- Lee, A. C. C., Cousens, S., Darmstadt, G. L., Blencowe, H., Pattinson, R., Moran, N. F., Hofmeyr, G. J., Haws, R. A., Bhutta, S. Z., & Lawn, J. E. (2011). Care during labor and birth for the prevention of intrapartum-related neonatal deaths: A systematic review and Delphi estimation of mortality effect [<https://doi.org/10.1186/1471-2458-11-S3-S10>](https://doi.org/10.1186/1471-2458-11-S3-S10). *BMC Public Health*, 11(Suppl. 3), S10.
- Lee, H., Maffioli, E. M., Veliz, P. T., Munro-Kramer, M. L., Phiri, T. K., Sakala, I., Kaunda, J., Chiboola, N. M., & Lori, J. R. (2021). Role of Savings and Internal Lending Communities (SILCs) in improving household wealth and financial

preparedness for birth in rural Zambia <https://doi.org/10.1093/heapol/czab049>. *Health Policy and Planning*, 36(8), 1269–1278.

Leonard, S. A., Main, E. K., Scott, K. A., Profit, J., & Carmichael, S. L. (2019). Racial and ethnic disparities in severe maternal morbidity prevalence and trends <https://doi.org/10.1016/j.annepidem.2019.02.007>. *Annals of Epidemiology*, 33, 30–36.

Lucas, D. N., & Bamber, J. H. (2021). Pandemics and maternal health: The indirect effects of COVID-19 <https://doi.org/10.1111/anae.15408>. *Anaesthesia*, 76(Suppl. 4), 69–75.

Lusti-Narasimhan, M., Khosla, R., Temmerman, M., & Young, H. B. (2014). Advancing the sexual and reproductive health and human rights of women beyond 2015 <https://doi.org/10.1111/1471-0528.13139>. *BJOG: An International Journal of Obstetrics & Gynaecology*, 121, 1–2.

Mackin, S. T., Nelson, S. M., Kerssens, J. J., Wood, R., Wild, S., Colhoun, H. M., Leese, G. P., Philip, S., & Lindsay, R. S. (2018). Diabetes and pregnancy: National trends over a 15 year period <https://doi.org/10.1007/s00125-017-4529-3>. *Diabetologia*, 61, 1081–1088.

Mackin, S. T., Nelson, S. M., Wild, S. H., Colhoun, H. M., Wood, R., & Lindsay, R. S. (2019). Factors associated with stillbirth in women with diabetes <https://doi.org/10.1007/s00125-019-4943-9>. *Diabetologia*, 62(10), 1938–1947.

Mamba, K. C., Muula, A. S., & Stones, W. (2017). Facility-imposed barriers to early utilization of focused antenatal care services in Mangochi District, Malawi—A mixed methods assessment <https://doi.org/10.1186/s12884-017-1631-y>. *BMC Pregnancy and Childbirth*, 17(1), 444.

Manguro, G., & Temmerman, M. (2022). A critical review of adolescent sexual and reproductive health and rights in Kenya <https://doi.org/10.1016/j.medj.2022.05.007>. *Medicine*, 3(6), 364–368.

Markus, H. R., & Kitayama, S. (2003). Models of agency: Sociocultural diversity in the construction of action. *Nebraska Symposium on Motivation*, 49, 1–57.

Mastroiacovo, P., & Leoncini, E. (2011). More folic acid, the five questions: Why, who, when, how much, and how <https://doi.org/10.1002/biof.172>. *BioFactors*, 37(4), 272–279.

Mchenga, M., Burger, R., & von Fintel, D. (2019). Examining the impact of WHO's Focused Antenatal Care policy on early access, underutilisation and quality of antenatal care services in Malawi: A retrospective study <https://doi.org/10.1186/s12913-019-4130-1>. *BMC Health Services Research*, 19(1), 295.

Miedema, S. S., Haardörfer, R., Girard, A. W., & Yount, K. M. (2018). Women's empowerment in East Africa: Development of a cross-country comparable measure <https://doi.org/10.1016/j.worlddev.2018.05.031>. *World Development*, 110, 453–464.

Miller, S., Abalos, E., Chamillard, M., Ciapponi, A., Colaci, D., Comandé, D., Diaz, V., Geller, S., Hanson, C., Langer, A., Manuelli, V., Millar, K., Morhason-Bello, I., Castro, C. P., Pileggi, V. N., Robinson, N., Skaer, M., Souza, J. P., Vogel, J. P., & Althabe, F. (2016). Beyond too little, too late and too much, too soon: A pathway towards evidence-based, respectful maternity care worldwide [https://doi.org/10.1016/S0140-6736\(16\)31472-6](https://doi.org/10.1016/S0140-6736(16)31472-6). *The Lancet*, 388(10056), 2176–2192.

- Montagu, D., Sudhinaraset, M., Diamond-Smith, N., Campbell, O., Gabrysch, S., Freedman, L., Kruk, M. E., & Donnay, F. (2017). Where women go to deliver: Understanding the changing landscape of childbirth in Africa and Asia [_<https://doi.org/10.1093/heapol/czx060>_](https://doi.org/10.1093/heapol/czx060). *Health Policy and Planning*, 32(8), 1146–1152.
- Moore Simas, T. A., Huang, M.-Y., Patton, C., Reinhart, M., Chawla, A. J., Clemson, C., & Eldar-Lissai, A. (2019). The humanistic burden of postpartum depression: A systematic literature review [_<https://doi.org/10.1080/03007995.2018.1552039>_](https://doi.org/10.1080/03007995.2018.1552039). *Current Medical Research and Opinion*, 35(3), 383–393.
- Moyer, C. A., Aborigo, R. A., Logonia, G., Affah, G., Rominski, S., Adongo, P. B., Williams, J., Hodgson, A., & Engmann, C. (2012). Clean delivery practices in rural northern Ghana: A qualitative study of community and provider knowledge, attitudes, and beliefs [_<https://doi.org/10.1186/1471-2393-12-50>_](https://doi.org/10.1186/1471-2393-12-50). *BMC Pregnancy and Childbirth*, 12, 1–14.
- Munabi-Babigumira, S., Lewin, S., Glenton, C., Velez, M., Gonçalves-Bradley, D. C., & Bohren, M. A. (2021). Factors that influence the provision of postnatal care by health workers: A qualitative evidence synthesis [_<https://doi.org/10.1002/14651858.CD014790>_](https://doi.org/10.1002/14651858.CD014790). *Cochrane Database of Systematic Reviews*, 7, i-14.
- National Institute for Health and Care Excellence. (2021). *Postnatal care* [_<https://www.nice.org.uk/guidance/ng194>_](https://www.nice.org.uk/guidance/ng194).
- Nieuwenhuijze, M., & Leahy-Warren, P. (2019). Women’s empowerment in pregnancy and childbirth: A concept analysis [_<https://doi.org/10.1016/j.midw.2019.07.015>_](https://doi.org/10.1016/j.midw.2019.07.015). *Midwifery*, 78, 1–7.
- Norheim, O. F., Jha, P., Admasu, K., Godal, T., Hum, R. J., Kruk, M. E., Gómez-Dantés, O., Mathers, C. D., Pan, H., Sepúlveda, J., Suraweera, W., Verguet, S., Woldemariam, A. T., Yamey, G., Jamison, D. T., & Peto, R. (2015). Avoiding 40% of the premature deaths in each country, 2010–30: Review of national mortality trends to help quantify the UN Sustainable Development Goal for health [_<https://doi.org/10.1016/S0140-6736\(14\)61591-9>_](https://doi.org/10.1016/S0140-6736(14)61591-9). *The Lancet*, 385(9964), 239–252.
- Nour, N. M. (2014). Global women’s health: Progress toward reducing sex-based health disparities [_<https://doi.org/10.1373/clinchem.2013.213181>_](https://doi.org/10.1373/clinchem.2013.213181). *Clinical Chemistry*, 60(1), 147–150.
- Obse, A. G., & Ataguba, J. E. (2021). Explaining socioeconomic disparities and gaps in the use of antenatal care services in 36 countries in sub-Saharan Africa [_<https://doi.org/10.1093/heapol/czab036>_](https://doi.org/10.1093/heapol/czab036). *Health Policy and Planning*, 36(5), 651–661.
- Okonofua, F. E. (2005). Achieving the Millennium Development Goals in Africa: How realistic? [Réalisation des Objectifs du Développement du Millénium en Afrique: Combien est-il Réaliste?] [_<https://doi.org/10.2307/3583407>_](https://doi.org/10.2307/3583407). *African Journal of Reproductive Health*, 9(3), 7.
- Oladapo, O. T., Diaz, V., Bonet, M., Abalos, E., Thwin, S. S., Souza, H., Perdoná, G., Souza, J. P., & Gülmezoglu, A. M. (2018). Cervical dilatation patterns of “low-risk” women with spontaneous labour and normal perinatal outcomes: A systematic review [_<https://doi.org/10.1111/1471-0528.14930>_](https://doi.org/10.1111/1471-0528.14930). *BJOG: An International Journal of Obstetrics and Gynaecology*, 125(8), 944–954.
- Oladapo, O. T., Tunçalp, Ö., Bonet, M., Lawrie, T., Portela, A., Downe, S., & Gülmezoglu, A. (2018). WHO model of intrapartum care for a positive childbirth experience: Transforming care of women and babies for improved health and wellbeing [_<https://doi.org/10.1111/1471-0528.15237>_](https://doi.org/10.1111/1471-0528.15237). *BJOG: An International Journal of Obstetrics & Gynaecology*, 125(8), 918–922.

- Osamor, P. E., & Grady, C. (2016). Women's autonomy in health care decision-making in developing countries: A synthesis of the literature <https://doi.org/10.2147/ijwh.s105483>. *International Journal of Women's Health*, 8, 191–202.
- Ota, E., da Silva Lopes, K., Middleton, P., Flenady, V., Wariki, W. M., Rahman, M. O., Tobe-Gai, R., & Mori, R. (2020). Antenatal interventions for preventing stillbirth, fetal loss and perinatal death: An overview of Cochrane systematic reviews <https://doi.org/10.1002/14651858.CD009599.pub2>. *The Cochrane Database of Systematic Reviews*, 2020(12), CD009599.
- Overbeck, G., Rasmussen, I. S., Siersma, V., Kragstrup, J., Ertmann, R. K., & Wilson, P. (2022). Mental well-being during stages of COVID-19 lockdown among pregnant women and new mothers <https://doi.org/10.1186/s12884-021-04374-4>. *BMC Pregnancy and Childbirth*, 22(1), 1–7.
- Palmer, K. R., Tanner, M., Davies-Tuck, M., Rindt, A., Papacostas, K., Giles, M. L., Brown, K., Diamandis, H., Fradkin, R., Stewart, A. E., Rolnik, D. L., Stripp, A., Wallace, E. M., Mol, B. W., & Hodges, R. J. (2021). Widespread implementation of a low-cost telehealth service in the delivery of antenatal care during the COVID-19 pandemic: An interrupted time-series analysis [https://doi.org/10.1016/S0140-6736\(21\)00668-1](https://doi.org/10.1016/S0140-6736(21)00668-1). *The Lancet*, 398(10294), 41–52.
- Pitt, C., Bath, D., Binyaruka, P., Borghi, J., & Martinez-Alvarez, M. (2021). Falling aid for reproductive, maternal, newborn and child health in the lead-up to the COVID-19 pandemic <https://doi.org/10.1136/bmjgh-2021-006089>. *BMJ Global Health*, 6(6), e006089.
- Poku, O. B., Ho-Foster, A. R., Entaile, P., Misra, S., Mehta, H., Rampa, S., Goodman, M., Arscott-Mills, T., Eschliman, E., Jackson, V., Melese, T., Becker, T. D., Eisenberg, M., Link, B., Go, V., Opondo, P. R., Blank, M. B., & Yang, L. H. (2020). “Mothers moving towards empowerment” intervention to reduce stigma and improve treatment adherence in pregnant women living with HIV in Botswana: Study protocol for a pragmatic clinical trial <https://doi.org/10.1186/s13063-020-04676-6>. *Trials*, 21(1), 832.
- Population Council. (1995). *Program of action of the 1994 international conference on population and development (chapters I–VIII)* <https://www.jstor.org/stable/2137429>. Wiley.
- Potts, M. (1990). Switzerland (WHO): Safe motherhood [https://doi.org/10.1016/0140-6736\(90\)91393-O](https://doi.org/10.1016/0140-6736(90)91393-O). *The Lancet*, 335(8705), 1580.
- Pradhan, M. R. (2022). Contraceptive mix and informed choice [https://doi.org/10.1016/S0140-6736\(22\)01282-X](https://doi.org/10.1016/S0140-6736(22)01282-X). *The Lancet*, 400(10348), 255–257.
- Prata, N., Fraser, A., Huchko, M. J., Gipson, J. D., Withers, M., Lewis, S., Ciaraldi, E. J., & Upadhyay, U. D. (2017). Women's empowerment and family planning: A review of the literature <https://doi.org/10.1017/S0021932016000663>. *Journal of Biosocial Science*, 49(6), 713–743.
- Pratley, P. (2016). Associations between quantitative measures of women's empowerment and access to care and health status for mothers and their children: A systematic review of evidence from the developing world <https://doi.org/10.1016/j.socscimed.2016.08.001>. *Social Science & Medicine*, 169, 119–131.
- Pugh, S. (2019). Politics, power, and sexual and reproductive health and rights: Impacts and opportunities <https://doi.org/10.1080/26410397.2019.1662616>. *Sexual and Reproductive Health Matters*, 27(2), 1–5.

- Rahman, M. M., Haque, S. E., & Zahan, M. S. (2011). Factors affecting the utilisation of postpartum care among young mothers in Bangladesh <https://doi.org/10.1111/j.1365-2524.2010.00953.x>. *Health & Social Care in the Community*, 19(2), 138–147.
- Renfrew, M. J., McFadden, A., Bastos, M. H., Campbell, J., Channon, A. A., Cheung, N. F., Silva, D. R. A. D., Downe, S., Kennedy, H. P., Malata, A., McCormick, F., Wick, L., & Declercq, E. (2014). Midwifery and quality care: Findings from a new evidence-informed framework for maternal and newborn care [https://doi.org/10.1016/S0140-6736\(14\)60789-3](https://doi.org/10.1016/S0140-6736(14)60789-3). *The Lancet*, 384(9948), 1129–1145.
- Requejo, J., Diaz, T., Park, L., Chou, D., Choudhury, A., Guthold, R., Jackson, D., Moller, A.-B., Monet, J.-P., Moran, A. C., Say, L., Strong, K. L., & Banerjee, A. (2020). Assessing coverage of interventions for reproductive, maternal, newborn, child, and adolescent health and nutrition <https://doi.org/10.1136/bmj.l6915>. *BMJ*, 368, l6915.
- Richardson, K. K., & Shannon, M. T. (2012). STI screening and treatment in pregnancy <https://doi.org/10.1097/01.NPR.0000422203.95229.ee>. *The Nurse Practitioner*, 37(12), 30–37.
- Rodríguez, M. I., Say, L., & Temmerman, M. (2014). Family planning versus contraception: What's in a name? [https://doi.org/10.1016/S2214-109X\(13\)70177-3](https://doi.org/10.1016/S2214-109X(13)70177-3) *The Lancet Global Health*, 2(3), e131–e132.
- Sacks, E., Finlayson, K., Brizuela, V., Crossland, N., Ziegler, D., Sauvé, C., Langlois, É. V., Javadi, D., Downe, S., & Bonet, M. (2022). Factors that influence uptake of routine postnatal care: Findings on women's perspectives from a qualitative evidence synthesis <https://doi.org/10.1371/journal.pone.0270264>. *PLOS ONE*, 17(8), e0270264.
- Sacks, E., & Langlois, É. V. (2016). Postnatal care: Increasing coverage, equity, and quality [https://doi.org/10.1016/S2214-109X\(16\)30092-4](https://doi.org/10.1016/S2214-109X(16)30092-4). *The Lancet Global Health*, 4(7), e442–e443.
- Sahariah, S. A., Potdar, R. D., Gandhi, M., Kehoe, S. H., Brown, N., Sane, H., Coakley, P. J., Marley-Zagar, E., Chopra, H., Shivshankaran, D., Cox, V. A., Jackson, A. A., Margetts, B. M., & Fall, C. H. (2016). A daily snack containing leafy green vegetables, fruit, and milk before and during pregnancy prevents gestational diabetes in a randomized, controlled trial in Mumbai, India <https://doi.org/10.3945/jn.115.223461>. *The Journal of Nutrition*, 146(7), 1453s–1460s.
- Say, L., Chou, D., Gemmill, A., Tunçalp, Ö., Moller, A.-B., Daniels, J., Gülmezoglu, A. M., Temmerman, M., & Alkema, L. (2014). Global causes of maternal death: A WHO systematic analysis [https://doi.org/10.1016/S2214-109X\(14\)70227-X](https://doi.org/10.1016/S2214-109X(14)70227-X). *The Lancet Global Health*, 2(6), e323–e333.
- Schisterman, E. F., Silver, R. M., Leshner, L. L., Faraggi, D., Wactawski-Wende, J., Townsend, J. M., Lynch, A. M., Perkins, N. J., Mumford, S. L., & Galai, N. (2014). Preconception low-dose aspirin and pregnancy outcomes: Results from the EAGeR randomised trial [https://doi.org/10.1016/S0140-6736\(14\)60157-4](https://doi.org/10.1016/S0140-6736(14)60157-4). *The Lancet*, 384(9937), 29–36.
- Senarath, U., & Gunawardena, N. S. (2009). Women's autonomy in decision making for health care in South Asia <https://doi.org/10.1177/1010539509331590>. *Asia Pacific Journal of Public Health*, 21(2), 137–143.
- Sharma, G., Ying, W., & Silversides, C. K. (2021). The importance of cardiovascular risk assessment and pregnancy heart team in the management of cardiovascular disease in pregnancy <https://doi.org/10.1016/j.ccl.2020.09.002>. *Cardiology Clinics*, 39(1), 7–19.

Shaw, D., Guise, J.-M., Shah, N., Gemzell-Danielsson, K., Joseph, K., Levy, B., Wong, F., Woodd, S., & Main, E. K. (2016). Drivers of maternity care in high-income countries: Can health systems support woman-centred care? [https://doi.org/10.1016/S0140-6736\(16\)31527-6](https://doi.org/10.1016/S0140-6736(16)31527-6) *The Lancet*, 388(10057), 2282–2295.

Shiffman, J., & Smith, S. (2007). Generation of political priority for global health initiatives: A framework and case study of maternal mortality [https://doi.org/10.1016/S0140-6736\(07\)61579-7](https://doi.org/10.1016/S0140-6736(07)61579-7). *The Lancet*, 370(9595), 1370–1379.

Singh, A., Padmadas, S. S., Mishra, U. S., Pallikadavath, S., Johnson, F. A., & Matthews, Z. (2012). Socio-economic inequalities in the use of postnatal care in India <https://doi.org/10.1371/journal.pone.0037037>. *PLOS ONE*, 7(5), e37037.

Slomian, J., Honvo, G., Emonts, P., Reginster, J.-Y., & Bruyère, O. (2019). Consequences of maternal postpartum depression: A systematic review of maternal and infant outcomes <https://doi.org/10.1177/1745506519844044>. *Women's Health*, 15, 1745506519844044.

Smith, G. N. (2015). The Maternal Health Clinic: Improving women's cardiovascular health <https://doi.org/10.1053/j.semperi.2015.05.012>. *Seminars in Perinatology*, 39(4), 316–319.

Smith, G. N., Pudwell, J., & Roddy, M. (2013). The Maternal Health Clinic: A new window of opportunity for early heart disease risk screening and intervention for women with pregnancy complications [https://doi.org/10.1016/S1701-2163\(15\)30841-0](https://doi.org/10.1016/S1701-2163(15)30841-0). *Journal of Obstetrics and Gynaecology Canada*, 35(9), 831–839.

Sommer, U., & Forman-Rabinovici, A. (2019). *Producing reproductive rights: Determining abortion policy worldwide* <https://doi.org/10.1017/9781108694407> (1st ed.). Cambridge University Press.

Souza, J., Tunçalp, Ö., Vogel, J., Bohren, M., Widmer, M., Oladapo, O., Say, L., Gülmezoglu, A., & Temmerman, M. (2014). Obstetric transition: The pathway towards ending preventable maternal deaths <https://doi.org/10.1111/1471-0528.12735>. *BJOG: An International Journal of Obstetrics & Gynaecology*, 121(S1), 1–4.

Starrs, A. M., Ezeh, A. C., Barker, G., Basu, A., Bertrand, J. T., Blum, R., Coll-Seck, A. M., Grover, A., Laski, L., Roa, M., Sathar, Z. A., Say, L., Serour, G. I., Singh, S., Stenberg, K., Temmerman, M., Biddlecom, A., Popinchalk, A., Summers, C., & Ashford, L. S. (2018). Accelerate progress—Sexual and reproductive health and rights for all: Report of the Guttmacher—Lancet Commission [https://doi.org/10.1016/S0140-6736\(18\)30293-9](https://doi.org/10.1016/S0140-6736(18)30293-9). *The Lancet*, 391(10140), 2642–2692.

Stein, A., Pearson, R. M., Goodman, S. H., Rapa, E., Rahman, A., McCallum, M., Howard, L. M., & Pariante, C. M. (2014). Effects of perinatal mental disorders on the fetus and child [https://doi.org/10.1016/S0140-6736\(14\)61277-0](https://doi.org/10.1016/S0140-6736(14)61277-0). *The Lancet*, 384(9956), 1800–1819.

Stephenson, J., Heslehurst, N., Hall, J., Schoenaker, D. A. J. M., Hutchinson, J., Cade, J. E., Poston, L., Barrett, G., Crozier, S. R., Barker, M., Kumaran, K., Yajnik, C. S., Baird, J., & Mishra, G. D. (2018). Before the beginning: Nutrition and lifestyle in the preconception period and its importance for future health [https://doi.org/10.1016/S0140-6736\(18\)30311-8](https://doi.org/10.1016/S0140-6736(18)30311-8). *The Lancet*, 391(10132), 1830–1841.

Stover, J., & Sonneveldt, E. (2017). Progress toward the goals of FP2020. *Studies in Family Planning*, 48(1), 83–88.

Sustainable Development Goals. (2022). *Sustainable Development Goals report* <https://www.un.org/sustainabledevelopment/progress-report/>.

Temmerman, M. (2016). From millennium to Sustainable Development Goals: Research and innovation for maternal and newborn health <<https://doi.org/10.1016/j.ejogrb.2016.07.024>>. *European Journal of Obstetrics & Gynecology and Reproductive Biology*, 206, 258.

Temmerman, M. (2019). Missed opportunities in women's health: Post-abortion care <[https://doi.org/10.1016/S2214-109X\(18\)30542-4](https://doi.org/10.1016/S2214-109X(18)30542-4)>. *The Lancet Global Health*, 7(1), e12–e13.

Temmerman, M., Khosla, R., Bhutta, Z. A., & Bustreo, F. (2015). Towards a new global strategy for women's, children's and adolescents' health <<https://doi.org/10.1136/bmj.h4414>>. *BMJ*, 351, h4414.

Temmerman, M., Khosla, R., & Say, L. (2014). Sexual and reproductive health and rights: A global development, health, and human rights priority <[https://doi.org/10.1016/S0140-6736\(14\)61190-9](https://doi.org/10.1016/S0140-6736(14)61190-9)>. *The Lancet*, 384(9941), e30–e31.

Director Early Child Development, World Health Organisation. (n.d.). *The global strategy for women's, children's and adolescents' health (2016–2030)* <[https://www.who.int/publications/i/item/the-global-strategy-for-women-s-children-s-and-adolescents-health-\(2016-2030\)-early-childhood-development-report-by-the-director-general](https://www.who.int/publications/i/item/the-global-strategy-for-women-s-children-s-and-adolescents-health-(2016-2030)-early-childhood-development-report-by-the-director-general)>.

Thompson, J., Undie, C.-C., Amin, A., Johnson, B. R., Khosla, R., Ouedraogo, L., Nkurunziza, T., Rich, S., Westley, E., Garcia, M., Birungi, H., & Askew, I. (2018). Correction to: Harmonizing national abortion and pregnancy prevention laws and policies for sexual violence survivors with the Maputo Protocol: Proceedings of a 2016 regional technical meeting in sub-Saharan Africa <<https://doi.org/10.1186/s12919-018-0103-3>>. *BMC Proceedings*, 12(1), 7.

Tunçalp, Ö., Pena-Rosas, J., Lawrie, T., Bucagu, M., Oladapo, O., Portela, A., & Metin Gülmezoglu, A. (2017). WHO recommendations on antenatal care for a positive pregnancy experience-going beyond survival <<https://doi.org/10.1111/1471-0528.14599>>. *BJOG: An International Journal of Obstetrics & Gynaecology*, 124(6), 860–862.

Tunçalp, Ö., Were, W., MacLennan, C., Oladapo, O., Gülmezoglu, A., Bahl, R., Daelmans, B., Mathai, M., Say, L., Kristensen, F., Temmerman, M., & Bustreo, F. (2015). Quality of care for pregnant women and newborns—The WHO vision <<https://doi.org/10.1111/1471-0528.13451>>. *BJOG: An International Journal of Obstetrics & Gynaecology*, 122(8), 1045–1049.

UN Women. (2015). *The Beijing Platform for Action: Inspiration then and now* <<https://beijing20.unwomen.org/en/about>>.

United Nations Department of Economic and Social Affairs (UN-DESA). (2015). *Global Sustainable Development Report, 2015 edition* <<https://www.un.org/en/development/desa/publications/global-sustainable-development-report-2015-edition.html>>.

UNICEF. (2020). *Ending preventable newborn deaths and stillbirths by 2030* <<https://data.unicef.org/resources/ending-preventable-newborn-deaths-and-stillbirths-by-2030/>>.

UNICEF, World Health Organization. (2017). *Tracking progress towards universal coverage for reproductive, newborn and child health: The 2017 report* <<http://countdown2030.org/pdf/Countdown-2030-complete-with-profiles.pdf>>.

United Nations. (2015). *Transforming our world: The 2030 agenda for sustainable development* [A/RES/70/1].

United Nations Development Programme. (2015). *The Millennium Development Goals Report 2015* <<https://www.undp.org/publications/millennium-development-goals-report-2015>>.

Villar, J., Ba'aqeel, H., Piaggio, G., Lumbiganon, P., Belizán, J. M., Farnot, U., Al-Mazrou, Y., Carroli, G., Pinol, A., Donner, A., Langer, A., Nigenda, G., Mugford, M., Fox-Rushby, J., Hutton, G., Bergsjø, P., Bakketeig, L., & Berendes, H. (2001). WHO antenatal care randomised trial for the evaluation of a new model of routine antenatal care <[https://doi.org/10.1016/S0140-6736\(00\)04722-X](https://doi.org/10.1016/S0140-6736(00)04722-X)>. *The Lancet*, 357(9268), 1551–1564.

Villar, J., Valladares, E., Wojdyla, D., Zavaleta, N., Carroli, G., Velazco, A., Shah, A., Campodónico, L., Bataglia, V., Faundes, A., Langer, A., Narváez, A., Donner, A., Romero, M., Reynoso, S., de Pádua, K. S., Giordano, D., Kubickas, M., Acosta, A., & WHO 2005 Global Survey on Maternal and Perinatal Health Research Group. (2006). Caesarean delivery rates and pregnancy outcomes: The 2005 WHO global survey on maternal and perinatal health in Latin America <[https://doi.org/10.1016/S0140-6736\(06\)68704-7](https://doi.org/10.1016/S0140-6736(06)68704-7)>. *The Lancet*, 367(9525), 1819–1829.

Vogel, J. P., Bohren, M. A., Tunçalp, Ö., Oladapo, O. T., & Gülmezoglu, A. M. (2016). Promoting respect and preventing mistreatment during childbirth <<https://doi.org/10.1111/1471-0528.13750>>. *BJOG: An International Journal of Obstetrics & Gynaecology*, 123(5), 671–674.

Vogel, J. P., Habib, N. A., Souza, J. P., Gülmezoglu, A. M., Dowswell, T., Carroli, G., Baaqeel, H. S., Lumbiganon, P., Piaggio, G., & Oladapo, O. T. (2013). Antenatal care packages with reduced visits and perinatal mortality: A secondary analysis of the WHO Antenatal Care Trial <<https://doi.org/10.1186/1742-4755-10-19>>. *Reproductive Health*, 10(1), 19.

Waage, J., Banerji, R., Campbell, O., Chirwa, E., Collender, G., Dieltiens, V., Dorward, A., Godfrey-Faussett, P., Hanvoravongchai, P., Kingdon, G., Little, A., Mills, A., Mulholland, K., Mwinga, A., North, A., Patcharanarumol, W., Poulton, C., Tangcharoensathien, V., & Unterhalter, E. (2010). The Millennium Development Goals: A cross-sectoral analysis and principles for goal setting after 2015 <[https://doi.org/10.1016/S0140-6736\(10\)61196-8](https://doi.org/10.1016/S0140-6736(10)61196-8)>. *The Lancet*, 376(9745), 991–1023.

Wahdan, M., Hakim, S., El Gaafary, M., Sos, D., Wassif, G., Hussein, W., Mokhtar, A., Hussein, A., El Awady, M., Rady, M., & Anwar, W. (2022). Rising trends in caesarean section in 6 Egyptian governorates <<https://doi.org/10.26719/emhj.22.012>>. *Eastern Mediterranean Health Journal* [La Revue De Sante De La Mediterranee Orientale], 28(5), 336–344.

Weeks, A., & Temmerman, M. (2016). New WHO antenatal care model—Quality worth paying for? <[https://doi.org/10.1016/S0140-6736\(16\)32233-4](https://doi.org/10.1016/S0140-6736(16)32233-4)> *The Lancet*, 388(10060), 2574–2575.

Weil, O., & Fernandez, H. (1999). Is safe motherhood an orphan initiative? <[https://doi.org/10.1016/S0140-6736\(99\)02369-7](https://doi.org/10.1016/S0140-6736(99)02369-7)> *The Lancet*, 354(9182), 940–943.

World Health Organization. (2006). *Quality of care: A process for making strategic choices in health systems* <<https://apps.who.int/iris/handle/10665/43470>>.

World Health Organization. (2014). *WHO recommendations on postnatal care of the mother and newborn*.

World Health Organization. (2016). *WHO recommendations on antenatal care for a positive pregnancy experience* <http://www.who.int/reproductivehealth/publications/maternal_perinatal_health/anc-positive-pregnancy-experience/en/> (p. 172).

World Health Organization. (2018a). *Global strategy for women's, children's and adolescents' health (2016–2030): Early childhood development* <[https://www.who.int/publications/i/item/the-global-strategy-for-women-s-children-s-and-adolescents-health-\(2016-2030\)-early-childhood-development-report-by-the-director-general](https://www.who.int/publications/i/item/the-global-strategy-for-women-s-children-s-and-adolescents-health-(2016-2030)-early-childhood-development-report-by-the-director-general)> [A71/19 Rev.1].

World Health Organization. (2018b). *Millennium Development Goals (MDGs)* <[https://www.who.int/news-room/fact-sheets/detail/millennium-development-goals-\(mdgs\)](https://www.who.int/news-room/fact-sheets/detail/millennium-development-goals-(mdgs))>.

World Health Organization. (2018c). *WHO recommendations: Intrapartum care for a positive childbirth experience* <<https://www.bing.com/search?q=IntrapartumCare+for+a+Positive+Childbirth+Experience&cvid=5c72448717a64319a9edf92a8b354030&aqs=edge..69i57j69i64.942j0j1&pglt=41&FORM=ANNTA1&PC=W037&ntref=1>> (p. 210).

World Health Organization. (2019). *Maternal mortality* <<https://www.who.int/news-room/fact-sheets/detail/maternal-mortality>>.

World Health Organization. (2022a). *Global Health Observatory: Births attended by skilled health personnel (%)* <[https://www.who.int/data/gho/data/indicators/indicator-details/GHO/births-attended-by-skilled-health-personnel-\(-\)](https://www.who.int/data/gho/data/indicators/indicator-details/GHO/births-attended-by-skilled-health-personnel-(-))>.

World Health Organization. (2022b). *Maternal health* <<https://www.who.int/health-topics/maternal-health>>.

World Health Organization. (2022c). *WHO recommendations on maternal and newborn care for a positive postnatal experience* <<https://www.who.int/activities/raising-the-importance-of-postnatal-care>> (p. 242).

World Health Organization, United Nations Children's Fund, World Bank Group. (2018). *Nurturing care for early childhood development: A framework for helping children survive and thrive to transform health and human potential* <<https://apps.who.int/iris/bitstream/handle/10665/272603/9789241514064-eng.pdf?ua=1>>.

Wu, H., Sun, W., Huang, X., Yu, S., Wang, H., Bi, X., Sheng, J., Chen, S., Akinwunmi, B., Zhang, C. J. P., & Ming, W.-K. (2020). Online antenatal care during the COVID-19 pandemic: Opportunities and challenges <<https://doi.org/10.2196/19916>>. *Journal of Medical Internet Research*, 22(7), e19916.

Yaya, S., Uthman, O. A., Ekholuenetale, M., & Bishwajit, G. (2018). Women empowerment as an enabling factor of contraceptive use in sub-Saharan Africa: A multilevel analysis of cross-sectional surveys of 32 countries <<https://doi.org/10.1186/s12978-018-0658-5>>. *Reproductive Health*, 15(1), 214.

Zanconato, G., Msolomba, R., Guarenti, L., & Franchi, M. (2006). Antenatal care in developing countries: The need for a tailored model <<https://doi.org/10.1016/j.siny.2005.10.002>>. *Seminars in Fetal and Neonatal Medicine*, 11(1), 15–20.

Related Articles

Community-Oriented Primary Health Care for Improving Maternal, Newborn, and Child Health

How Primary Maternal-Child Healthcare Reduces Mortality among Mothers and Children throughout the World: A Historical and Personal Reflection on Progress and Missed Opportunities